

Functional Analysis

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1 Background Material

Definition 1.1. (Cardinality) Let $f : X \rightarrow Y$ be a bijection, if $|X| = |Y|$ then X, Y have same Cardinality. We also define

1. $|X| \leq |Y|$ as smaller or equal
2. $|X| < |Y|$ as strictly smaller

Theorem 1.2. (Cantor Bernstein Schroeder Theorem) $|X| \leq |Y|$ and $|Y| \leq |X| \implies |X| = |Y|$.

Example. Define $X = (0, 1)$, $Y = [0, 1]$, and $\mathbb{R}$, it suffices to show

$$|X| \leq |Y| \leq |\mathbb{R}| \leq |X|$$

To show $|\mathbb{R}| \leq |X|$, define $f : \mathbb{R} \rightarrow (0, 1)$ by

$$f(x) = \frac{1}{\pi} \arctan x + \frac{1}{2}$$

Definition 1.3.

1. We say X is finite if $|X| = |\{1, \dots, n\}|$ for some n .
2. We say X is countable if $|X| = |\mathbb{N}| = \aleph_0$.
3. We say X has cardinality of continuum if $|X| = |\mathbb{R}| = \aleph$.

Examples.

1. $\mathbb{Z}$ is countable.
2. $\mathbb{Q}$ is countable.
3. Every countable union of countably many sets is countable.

Theorem 1.4. $\mathbb{R}$ is uncountable.

Proof. This can be done by contradiction. We define

$$x_1 = 0.a_{11}a_{12}\dots$$

$$x_2 = 0.a_{21}a_{22}\dots$$

$$x_3 = 0.a_{31}a_{32}\dots$$

etc. Now define

$$y = 0.b_1b_2\dots$$

where $b_1 \neq a_{11}, b_2 \neq a_{22}, \dots$. Then y does not belong to any of $\{x_n\}$ so $\mathbb{R}$ is uncountable. $\square$

Definition 1.5. (Metric Spaces) A metric space (X, ρ) is a set X together with a function $\rho : X \times X \rightarrow \mathbb{R}_{\geq 0}$ satisfies

1. $\rho(x, y) = 0$ if and only if $x = y$
2. $\rho(x, y) = \rho(y, x)$
3. $\rho(x, y) \leq \rho(x, z) + \rho(z, y)$

Definition 1.6. Any $A \subset \mathbb{R}^n$ metric space, define the topology

$$B(x, r) = \{y \in X : \rho(x, y) < r\}$$

$$B[x, r] = \{y \in X : \rho(x, y) \leq r\}$$

Note that $B_r(x)$ is another notation for $B(x, r)$.

Definition 1.7. We say $U \subseteq X$ is open if it contains an open ball about each of its points. That is, for all $x \in U$, there exists $\epsilon > 0$ such that $B_\epsilon(x) \subseteq U$. A set $A \subseteq X$ is closed if its complement is open.

Definition 1.8. Suppose that $x \in U$ and U is open in X . Then U is a neighborhood of x .

Proposition 1.9. Arbitrary union of open sets is open, finite intersection of open sets is open. Arbitrary intersection of closed sets is closed, finite union of closed sets is closed.

Proof.

1. We first show the union of any collection of open subsets of X is open. Suppose $\{A_j : j \in J\}$ be a collection of

open sets, indexed by J and let $A = \bigcup_{j \in J} A_j$. Let $x \in A$ be arbitrary then x belongs to one of the the sets A_j . Since this set is open, it contains an open ball about x ; clearly this ball lies in A thus, A meets the definition of an open set.

2. Then we show that finite intersection of open sets is open. Suppose $A_1, A_2, \dots, A_n$ are open sets and let $A = \bigcap_{i=1}^n A_i$. Let $x \in A$ be arbitrary. Then $x \in A_i$ for each $i = 1, 2, \dots, n$. For each i since A_i is open, there exists $r_i > 0$ such that $B_{r_i}(x) \subseteq A_i$ for each $i = 1, 2, \dots, n$. Let $r = \min\{r_1, r_2, \dots, r_n\}$. Then $B_r(x) \subseteq B_{r_i}(x) \subseteq A_i$ for each $i = 1, 2, \dots, n$ and hence $B_r(x) \subseteq A$. But $x \in A$ is chosen arbitrarily so by definition A is an open set.

3. The statement for closed sets can be proved similarly through de Morgan laws.

□

Example. Intersection of open sets is not necessarily open:

$$\bigcap_{n=1}^{\infty} (-1, \frac{1}{n}) = (-1, 0]$$

Definition 1.10. $A \subset X$ metric space bounded if $A \subset B(x, r)$ for some (x, r) .

Definition 1.11. $\{x_n\}_{n=1}^{\infty} \subset X$, we say

$$x_n \rightarrow x \quad \text{or} \quad \lim_{n \rightarrow \infty} x_n = x$$

if $\rho(x, x_n) \rightarrow 0$.

Proposition 1.12. $F \subset X$ is closed if and only if $\forall x_n \subset F$ such that $x_n \rightarrow x$ then $x \in F$.

Proof.

($\implies$) By contradiction, suppose $\exists x_n \rightarrow x$ such that $\{x_n\} \in F$ but $x \notin F$. Then $\forall r > 0$, $x_n \in B(x, r)$ for some n . F is closed so $x \in F^c$ is open so $\exists r$ such that $B(x, r) \in F^c$ i.e. $B(x, r) \cap F = \emptyset$.

($\impliedby$) Suppose F is not closed then $\exists x \in F^c$ such that $B(x, \frac{1}{n}) \cap F \neq \emptyset$. By definition, $\rho(x, x_n) \rightarrow 0$ but $x_n \rightarrow x \notin F$ so contradiction here. □

Definition 1.13. $A \subset X$, $x \in X$ be accumulation point of A if $\exists \{x_n\} \subset A$ such that $x_n \rightarrow x$. $a \in A$ is an interior point if $\exists r > 0$ such that $B(x, r) \subset A$.

Definition 1.14. Closure $\bar{A}$ of A is the union of all accumulation points of A and the interior $\text{int}(A)$ is the union of all interior points.

Proposition 1.15.

1. $\bar{A}$ is closed and $\bar{A}$ is the smallest closed set containing A
2. $\text{int}(A)$ is open and it is the biggest open set contained in A .

Proof. It suffices to show $\bar{A}$ is closed if and only if $\{x_n\} \subset \bar{A}$ such that $x_n \rightarrow x \implies x \in \bar{A}$. $\forall n, \exists A \ni x_{n,k} \rightarrow x_n$ take $k(n)$ such that $\rho(x_{n,k(n)}, x_n) < \frac{1}{n}$ then

$$\rho(x_{n,k(n)}, x) \leq \rho(x_{n,k(n)}, x_n) + \rho(x_n, x)$$

which implies $x_{n,k(n)} \rightarrow x$ so $x \in \bar{A}$. □

Definition 1.16. $A \subset X, x \in X$, define

$$\text{dist}(x, A) = \inf_{a \in A} \rho(x, a)$$

Proposition 1.17. $\text{dist}(x, A) = 0 \iff x \in \bar{A}$ and

$$\bar{A} = \{x \in X : \text{dist}(x, A) = 0\}$$

Proof.

($\implies$) $\text{dist}(x, A) = 0$ implies $\inf_{a \in A} \rho(x, a) = 0$. Suppose $x \notin \bar{A}$ i.e. $x \in X \setminus \bar{A}$ which is an open set. Thus $\exists \varepsilon > 0$ such that

$$B(x, \varepsilon) \setminus \{x\} = \{y \in X : 0 < d(x, y) < \varepsilon \subset X \setminus \bar{A}\}$$

But then

$$\text{dist}(x, A) = \inf_{a \in A} \rho(x, a) \geq \inf\{d(x, y) : y \in B(x, \varepsilon) \setminus \{x\}\} > 0$$

which is a contradiction.

($\Leftarrow$) Suppose $x \in \bar{A}$. If $x \in \text{int}(A)$ then $\inf\{d(x, a) : a \in A\} = d(x, x) = 0$. Now instead, if $x \notin \text{int}(A)$ then there exists a sequence $\{a_k\} \subset A$ such that a_k converges to x . Then given $\varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $\forall k \geq N$, $d(a_k, x) < \varepsilon$. In this case,

$$\text{dist} = \inf\{d(x, a_k)\} = \lim_{k \rightarrow \infty} d(x, a_k) = 0$$

□

Definition 1.18. (Complete Metric Spaces) $\{x_n\} \subset X$, $\{x_i\}$ is Cauchy sequence if $\forall \varepsilon > 0, \exists N$ such that $\forall n, m > N$, $\rho(x_n, x_m) < \varepsilon$. Convergent sequence are Cauchy.

Lemma 1.19. Say $\{x_n\}$ to be Cauchy. If $x_{n_k} \rightarrow x$ then $x_n \rightarrow x$.

Proof. Suppose that $\{x_n\}$ is a Cauchy sequence in (X, ρ) and that $\{x_{n_k}\}$ is a subsequence converging to $x \in X$. Let $\varepsilon > 0$, note that $n_k \geq k$ for all $k \geq 0$. Let N be such that $\forall n, m \geq N$, $\rho(x_n, x_m) < \varepsilon/2$. Let K be such that $\forall k \geq K$, $\rho(x_{n_k} - x) < \varepsilon/2$. Let $M = \max\{N, K\}$ then if $n \geq M$ we have

$$\rho(x_n, x) \leq \rho(x_n, x_{n_k}) + \rho(x_{n_k}, x) = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

□

Definition 1.20. X is complete if any Cauchy sequence in X converges to a point in X .

Examples. Complete spaces:

1. $\mathbb{R}^n, \mathbb{C}^n$
2. Closed subsets

Theorem 1.21. (Cantor's intersection theorem) The followings are equivalent:

1. X is complete
2. For any nested sequence

$$K_n = \{B[x_n, r_n]\}$$

where $K_1 \supset K_2 \supset \dots \supset K_n \supset \dots$ with $r_n \rightarrow 0$ has non-empty intersection $\bigcap K_n \neq \emptyset$.

Proof.

(1 $\Rightarrow$ 2) Idea: $\bigcap K_n \supset \lim_{n \rightarrow \infty} x_n$. Need to check $\{x_n\}$ Cauchy. $\forall \varepsilon > 0, \exists N$ such that $\forall n > N, r_n < \varepsilon$. $\forall n, m > N$, $x_n, x_m \in K_N = B[x_N, r_N]$ and

$$\rho(x_n, x_m) \leq \rho(x_n, x_N) + \rho(x_N, x_m) < r_N + r_N < 2\varepsilon$$

then it's Cauchy sequence.

(1 $\Leftarrow$ 2) Ideas: construct K_m such that $\bigcap K_m = \lim_{n \rightarrow \infty} x_n$. Take $\varepsilon = 1 \Rightarrow \exists N_1$ such that $\forall n, m \geq N_1$, $\rho(x_n, x_m) < \varepsilon$. Repeat the process with $\varepsilon = \frac{1}{2}$ then $\exists N_2$ such that $\forall n, m \geq N_2$, $\rho(x_n, x_m) < \varepsilon/2$. Take $\varepsilon = \frac{1}{4}$, $\exists N_1$ such that $\forall n, m > N_1$, $\rho(x_n, x_m) < \frac{1}{4}$. Take $K_1 = B[x_{N_1}, \frac{1}{2}]$ so that $\forall y \in B[x_{N_1}, \frac{1}{4}]$, $B(y, \frac{1}{4}) \subset K$. Take $\varepsilon = \frac{1}{16}$, $\exists N_2 > N_1$ such that $\forall n, m > N_2$, $\rho(x_n, x_m) < \frac{1}{16}$ so $K_2 = B[x_{N_2}, \frac{1}{16}]$ at step n we construct K_{m+1} such that $\forall n > N_m$, $\{x_n\}_{n > N_m} \subset K_m$ where $K_m = B[x_{N_m}, 2^{-2m+1}]$.

$$\Rightarrow \bigcap K_m \ni x$$

we claim that $\lim_{m \rightarrow \infty} x_{N_m} = x$.

□

Definition 1.22. $(X, \rho_x), (Y, \rho_y)$ are two metric spaces then $f : X \rightarrow Y$. We say that f is continuous at $x \in X$ if $\forall \varepsilon > 0, \exists \delta > 0$ such that

$$\begin{aligned} f^{-1}(B_Y(f(x), \varepsilon)) &\supset B_X(x, \delta) \\ \Leftrightarrow \rho_X(x_1, x) < \delta &\Rightarrow \rho_Y(f(x), f(x_1)) < \varepsilon \end{aligned}$$

f is continuous if it is continuous at all points of X

Proposition 1.23. $f : X \rightarrow Y$ is continuous if and only if open $U \subset Y$, $f^{-1}(U)$ is open in X .

Definition 1.24. $f : X \rightarrow Y$ is homeomorphism if f is continuous bijection & f^{-1} is continous.

Definition 1.25. $f : (X, \rho_X) \rightarrow (Y, \rho_Y)$

1. f is distance-preserving if $\forall x_1, x_2 \in X$,

$$\rho_Y(f(x_1), f(x_2)) = \rho_X(x_1, x_2)$$

2. f is isometry if f is bijection.

Definition 1.26. $(\tilde{X}, \tilde{\rho})$ as a completion of (X, ρ) if

1. $(\tilde{X}, \tilde{\rho})$ is complete
2. $\exists$ distance-preserving map $\psi : X \rightarrow \tilde{X}$
3. $\psi(X)$ is dense in $\tilde{X}$ i.e. $\phi(\tilde{X}) = \tilde{X}$

Theorem 1.27. $\forall (X, \rho)$, $\exists$ completion on $(\tilde{X}, \tilde{\rho})$. It is unique up to isomorphism.

Proof. If $(\tilde{X}_1, \tilde{\rho}_1), (\tilde{X}_2, \tilde{\rho}_2)$ are completions then $\exists \psi : \tilde{X}_1 \rightarrow \tilde{X}_2$ with ϕ isometry. We define $\tilde{X}$ to be the "set of Cauchy sequence with the same limit", $\{x_n\}, \{y_n\}$ two Cauchy sequence and $\{x_n\} \sim \{y_n\}$ if $\rho(x_n, y_n) \rightarrow 0$. Define $\tilde{X}$ be set of equivalence classes

$$\tilde{\rho}(\{x_n\}, \{y_n\}) = \lim_{n \rightarrow \infty} \rho(x_n, y_n)$$

□

Definition 1.28. (Compactness in metric spaces) $A \subset X$. A is compact if $\forall$ open covering $A \subset \bigcup_{j \in J} U_j$ then $\exists j_1, \dots, j_n \in J$ such that

$$A \subset \bigcup_{i=1}^n U_{j_i}$$

Proposition 1.29. Any compact set is closed.

Proof. $\forall y \in K$, we have $V_y = B(x, \frac{\rho(x, y)}{2})$ and $U_y = B(y, \frac{\rho(x, y)}{2})$ then

$$K \subset \bigcup_{y \in K} U_y \implies \exists y_1, \dots, y_n$$

such that $K \subset \bigcup_{i=1}^n U_{y_i} \cdot \bigcap_{i=1}^n V_{y_i}$ is open and it contains x disjoint from $\bigcup_{i=1}^n U_{y_i} \supset K$.

□

Proposition 1.30. Every closed subset of compact set is compact.

Proof. $K \supset F$ – closed and $F \subset \bigcup_{j \in J} U_j$. $K \subset K \setminus F \cup \bigcup_{j \in J} U_j$ is an open cover of K . Thus, $\exists$ finite subcover

$$K \subset K \setminus F \cup \bigcup_{i=1}^n U_{j_i}$$

$$F \subset \bigcup_{i=1}^n U_{j_i}$$

thus it's compact.

□

Proposition 1.31. X – compact if and only if for every family $\mathcal{F}$ of closed sets such that

$$\bigcap_{F \in \mathcal{F}} F = \emptyset$$

there exists $F_1, \dots, F_n \in \mathcal{F}$

$$\bigcap_{i=1}^n F_i = \emptyset$$

Proof. U – open if and only if U^c is closed and $\bigcup U_j = X$ if and only if $\bigcap U_j^c = \emptyset$. $\bigcup_{i=1}^n U_{j_i} = X$ if and only if $\bigcap U_{j_i}^c = \emptyset$.

□

Definition 1.32. X is totally bounded if $\forall \varepsilon > 0$, $\exists x_1, x_2, \dots, x_n \in X$ such that $X = \bigcup_{i=1}^n B(x_i, \varepsilon)$.

Theorem 1.33. The following are equivalent:

1. (X, ρ) – compact
2. $\forall F_1 \supset F_2 \supset \dots$ nested sequence of non-empty closed sets

$$\bigcap F_i \neq \emptyset$$

3. X is sequentially compact. $\forall \{x_n\} \subset X$, $\exists n_k$ such that $x_{n_k} \rightarrow x \in X$ as $k \rightarrow \infty$.
4. X is complete & totally bounded.

Proof.

(1 $\implies$ 2) If $\bigcap F_i = \emptyset \implies$ by finite intersection property

$$F_{\min_k i_k} = \bigcap_{k=1}^n F_{i_k} = \emptyset$$

(2 $\implies$ 3) $F_k = \overline{\{x_n\}_{n>k}}$

(3 $\implies$ 4) X is complete because $\{x_m\}$ Cauchy so $x_{n_k} \rightarrow x \implies x_n \rightarrow x$. Suppose X is not totally bounded then $\exists \varepsilon > 0$ such that $\forall x_1, \dots, x_n, \exists y \in X \setminus \bigcup_{i=1}^n B(x_i, \varepsilon)$. Take $x_1 \in X$ any point and $x_2 \in X \setminus B(x_1, \varepsilon)$ then $\rho(x_1, x_2) \geq \varepsilon$. Take $x_3 \in X \setminus (B(x_1, \varepsilon) \cup B(x_2, \varepsilon))$. We can repeat the process so $\rho(x_n, x_m) > \varepsilon$ which is not Cauchy a contradiction.

(4 $\implies$ 1) Suppose X is not compact $\implies$ there is a covering $\mathcal{O}$ finite subcover, $X = \bigcup_{i=1}^n B(y_i, 1)$ for some i , $\mathcal{O}$ does not have finite subcover of $B(y_i, 1)$. Denote its centre by y_1 . recursively $B(y_n, 2^{-n})$ has a finite cover by balls of radius 2^{-n-1} , and one of them has no finite subcover from $\mathcal{O}$, and we denote its centre by y_{n+1} . The sequence $\{y_n\}$ is a Cauchy since its tail belongs to balls of arbitrarily small radius; since X is complete $\{y_n\}$ converges to $y \in X$. Let $U \in \mathcal{O}$ be an open set of the cover containing y . There is $\varepsilon > 0$ such that $B(y, \varepsilon) \subset U$. For N large enough that for all $n \geq N$ we have $2^{-n} < \varepsilon/2$, we have that $B(y_n, 2^{-1}) \subset B(y, \varepsilon)$. But that means that $B(y_n, 2^{-1}) \subset U$, in contradiction with the fact that there is no finite subcover from $\mathcal{O}$. $\square$

Definition 1.34. (Pre-compact) $A \subset X$ is pre-compact if $\bar{A}$ is compact.

Proposition 1.35. If X is complete then $A \subset X$ is pre-compact if and only if it's totally bounded.

Proof. A is totally bounded if and only if $\bar{A}$ is totally bounded. Since $A \subset \bigcup B(x, \varepsilon) \implies \bar{A} \subset \bigcup B(x, \varepsilon) \subset \bigcup B(x, 2\varepsilon)$. If $B \subset X$ complete then B -complete if and only if B is closed. $\bar{A}$ is compact if and only if $\bar{A}$ is complete and totally bounded so $\bar{A}$ is totally bounded if and only if A is totally bounded. $\square$

Definition 1.36. $A \subset X$, A is nowhere dense if $\bar{A}$ has empty interior.

Proposition 1.37. The following are equivalent:

1. A is nowhere dense
2. $(\bar{A})^c$ is dense
3. Every open set U contains open $v \subset U$ such that $V \cap A = \emptyset$.

Proof. (1 $\implies$ 2) For any set,

$$(\text{int}(C))^c = \overline{(C^c)}$$

apply to $C = \bar{A}$.

(2 $\implies$ 3) Any open set $(\bar{A})^c \subset X$ is dense if and only if any open set in X contains $V \subset (\bar{A})^c$. $\square$

Example. $\mathbb{R} \subset \mathbb{R}^2$.

Theorem 1.38. (Baire category theorem) A complete metric space cannot be written as a countable union of nowhere dense sets.

Proof. Prove by Cantor intersection theorem, suppose $X = \bigcup_{i=1}^{\infty} A_i$ where A_i are nowhere dense. Let $K_0 = B[x_0, 1] \supset B(x_0, 1)$. We can choose $K_1 = B[x_1, r_1] \supset B(x_1, r_1)$ where $B(x_1, r_1) \cap A_1 = \emptyset$. say $r_1 \leq \frac{1}{2}$ then we construct a nested sequence such that

$$K_n \supset K_{n+1} \supset \dots$$

and $K_n \cap A_n = \emptyset$, $r_n \rightarrow 0$. By Cantors intersection theorem, $x \in \bigcap K_n$ and at the same time $x \notin A_n$ for all n which implies

$$x \notin \bigcup A_n$$

which contradicts our original assumption. $\square$

Remarks.

1. Baire called countable union of nowhere dense sets "sets of 1st category" rest are "2nd category".
2. 1st category $\rightarrow$ "meagre", their complements $\rightarrow$ "residual".
3. If some property is satisfied for all elements of a residual set $A \subset X$ in a complete metric space, then we say that this property is generic.

Example. Prove $X = [0, 1]$ is not countable. Suppose it's countable then it can be written as a countable union of single points or $X = \bigcup_{i=1}^{\infty} \{x_i\}$ since $\{x_i\}$ is nowhere dense so this contradicts Baire category theorem.

Example. One can also use Baire category to show incompleteness of certain space. We can show $\mathbb{Q} \subset \mathbb{R}$ is not complete.

2 Normed space

2.1 Basic definition

Definition 2.1. (Norms on a vector space). Let V be a vector space over a field $\mathbb{F}$. A norm $\|\cdot\|$ on V is a function from V to $\mathbb{R}_{\geq 0} = [0, +\infty)$ that satisfies the following three properties:

1. $\|u + v\| \leq \|u\| + \|v\|$ for all $u, v \in V$.
2. $\|\lambda u\| \leq |\lambda| \|u\|$ for all $u \in V$ and $\lambda \in \mathbb{F}$.
3. $\|u\| = 0$ if and only if $u = 0$.

Remark. $\rho(x, y) = \|x - y\|$.

Remark. X, Y metric spaces

$$\rho_{X \times Y}((x_1, y_1), (x_2, y_2)) = \max\{\rho_X(x_1, x_2), \rho_Y(y_1, y_2)\}$$

Same for norms

$$\|(x, y)\|_{X \times Y} = \max\{\|x\|_X, \|y\|_Y\}$$

Proposition 2.2. X is normed space. Then addition $+: X \times X \rightarrow X$ and multiplication $\times: \mathbb{F} \times X \rightarrow X$ are continuous.

Proof. For $(x_1, y_1) \in X \times X$ and $(x_2, y_2) \in X \times X$ then

$$\|(x_2 + y_2) - (x_1 + y_1)\|_X \leq$$

in terms of $\|(x_1, y_1) - (x_2, y_2)\|_{X \times X}$. And

$$\begin{aligned} \|(x_2 + y_2) - (x_1 + y_1)\|_X &\leq \|x_2 - x_1\|_X + \|y_2 - y_1\|_X \\ &\leq \max\{\|x_2 - x_1\|_X, \|y_2 - y_1\|_X\} \\ &= 2\|(x_1, y_1) - (x_2, y_2)\|_{X \times X} \end{aligned}$$

□

Definition 2.3. $W \subset X$ normed space, W is closed if it is closed respect to ρ . W is subspace then its closure $\bar{W}$ is a closed subspace.

Definition 2.4. X is a Banach space if it's normed and complete.

Proposition 2.5. X -normed space. The following are equivalent:

1. X is Banach
2. $\forall$ sequence $\{x_n\} \subset X$ such that

$$\sum_{n=1}^{\infty} \|x_n\| < \infty$$

the series $\sum_{n=1}^{\infty} x_n$ converges.

Proof. $(1 \implies 2)$ $A_k = \sum_{k=1}^n x_k$. It is enough to show A_k is Cauchy.

$$\|A_m - A_n\| = \left\| \sum_{k=n+1}^m x_k \right\| \leq \sum_{k=n+1}^m \|x_k\| \rightarrow 0$$

□

Examples. ℓ^p spaces: $X = \mathbb{R}^d$ or $\mathbb{C}^d$ and

$$\|x\|_p = \left(\sum_{i=1}^d |x_i|^p \right)^{1/p}$$

and $\|x\|_{\infty} = \max_{i \in [1, d]} |x_i|$

Definition 2.6. $p, q \geq 1$ Hölder conjugate if

$$\frac{1}{p} + \frac{1}{q} = 1$$

Lemma 2.7. (Young's inequality) $a, b \geq 0$, p, q conjugate then:

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

with inequality if and only if $a^p = b^q$.

Proof. Since e^x is a convex function, by Jensen's inequality we have

$$e^{x/p+y/q} \leq \frac{1}{p}e^x + \frac{1}{q}e^y$$

then if we let $x = \log |a|^p$ and $y = \log |b|^q$ we immediately have

$$\begin{aligned} e^{\log |ab|} &\leq \frac{1}{p}|a|^p + \frac{1}{q}|b|^q \\ \implies |ab| &\leq \frac{1}{p}|a|^p + \frac{1}{q}|b|^q \end{aligned}$$

□

Theorem 2.8. (Hölder inequality) Let $v \cdot w = \sum_{n=1}^N v_n w_n$ denote the Euclidean inner-product of vectors $v, w \in \mathbb{R}^n$. Show that

$$|v \cdot w| \leq \|v\|_p \|w\|_q,$$

where $\|v\|_p = (\sum_{n=1}^N |v_n|^p)^{1/p}$ and $\|w\|_q = (\sum_{n=1}^N |w_n|^q)^{1/q}$.

Proof. Define the following

$$\tilde{v} = \frac{v}{\|v\|_p}, \quad \tilde{w} = \frac{w}{\|w\|_q}$$

then we have $\|\tilde{v}\|_p = \|\tilde{w}\|_q = 1$. Now if we use the result from part(a) we have the following

$$\begin{aligned} |\tilde{v}_i \tilde{w}_i| &\leq \frac{1}{p}|\tilde{v}_i|^p + \frac{1}{q}|\tilde{w}_i|^q \\ \implies \sum_{i=1}^N |\tilde{v}_i \tilde{w}_i| &\leq \sum_{i=1}^N \left(\frac{1}{p}|\tilde{v}_i|^p + \frac{1}{q}|\tilde{w}_i|^q \right) \\ \implies |\tilde{v} \cdot \tilde{w}| &\leq \frac{\|\tilde{v}\|_p}{p} + \frac{\|\tilde{w}\|_q}{q} = 1 \end{aligned}$$

Now multiply each side of the inequality by $\|v\|_p$ and $\|w\|_q$ we immediately have

$$|v \cdot w| \leq \|v\|_p \|w\|_q$$

□

Theorem 2.9. (Minkowski inequality) $p \in [1, +\infty]$, then $\forall x, y \in \mathbb{R}^d$

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p$$

Proof. Let $p < \infty$, then

$$|a + b|^p \leq |a + b|^{p-1}(|a| + |b|)$$

so

$$\|x + y\|_p^p = \sum_i |x_i + y_i|^p \leq \sum_i |x_i + y_i|^{p-1}(|x_i| + |y_i|)$$

by Hölder,

$$\begin{aligned} \sum_i |x_i + y_i|^{p-1}|x_i| &\leq \|x\|_p \left(\sum_i |x_i + y_i|^{q(p-1)} \right)^{1/q} \\ &= \|x\|_p \|x + y\|_p^{p-1} \end{aligned}$$

thus,

$$\|x + y\|_p^p \leq \|x\|_p \|x + y\|_p^{p-1} + \|y\|_p \|x + y\|_p^{p-1} \implies \|x + y\|_p \leq \|x\|_p + \|y\|_p$$

□

$\mathbb{R}^d, \mathbb{C}^d$ have many norms $\|\cdot\|_p$.

Definition 2.10. $\|\cdot\|_1, \|\cdot\|_2$ on X normed spaces. Then they are equivalent if $\exists C_1, C_2 > 0$ such that $\forall x \in X$

$$C_1 \|x\|_1 \leq \|x\|_2 \leq C_2 \|x\|_1$$

Example. $X \times X$ where X normed space $\|\cdot\|$ and

$$\|(x, y)\|_1 = \max(\|x\|, \|y\|)$$

$$\|(x, y)\|_2 = \|x\| + \|y\|$$

$$\|(x, y)\|_3 = \sqrt{\|x\|^2 + \|y\|^2}$$

then all of them are equivalent. Since

$$\max(\|x\|, \|y\|) \leq \|x\| + \|y\| \leq 2 \max(\|x\|, \|y\|)$$

Theorem 2.11. On $\mathbb{R}^d, \mathbb{C}^d$ all norms are equivalent.

Proof. It is sufficient to show that $\|\cdot\|$ is equivalent to $\|\cdot\|_\infty$. Sufficient to prove for $\|x\|_\infty = 1$, need to show on $S = \{x : \|x\|_\infty = 1\}$ $\|\cdot\|$ is bounded. If $f(x) = \|x\|$ is continuous on $(S, \|\cdot\|_\infty)$. Let $x, y \in X$ with $x = (x_1, \dots, x_d)$, $y = (y_1, \dots, y_d)$ or $x = \sum x_i e_i$ then

$$\|x - y\| = \left\| \sum (x_i - y_i) e_i \right\| \leq \sum |x_i - y_i| \|e_i\| \leq \|x - y\|_\infty \sum_{i=1}^d \|e_i\|$$

which implies continuity. $\square$

Definition 2.12. X normed space. X is separable if it has a countable dense subset.

Example. $(\mathbb{R}, |\cdot|)$ is separable since $\mathbb{Q} \subset \mathbb{R}$. $(\mathbb{R}^n, |\cdot|)$ is separable since $\mathbb{Q}^n \subset \mathbb{R}^n$.

Proposition 2.13. $\mathbb{R}^d, \mathbb{C}^d$ with $\|\cdot\|$ are separable Banach spaces.

Proof. Completeness follows the fact that every Cauchy sequence in $\mathbb{R}^d, \mathbb{C}^d$ converge in $\|\cdot\|_\infty$. Separability follows from previous example. $\square$

Corollary 2.14. All finite dimensional subspaces of a normed space are closed.

Definition 2.15. ℓ^p is the space containing all $\{x_n\}_{n=1}^\infty$ satisfies

$$\|x\|_p = \left(\sum_{k=1}^\infty |x_k|^p \right)^{1/p} < \infty$$

Theorem 2.16. (Hölder inequality for sequence) Define

$$x = (x_1, x_2, \dots)$$

$$y = (y_1, y_2, \dots)$$

$$z = x \cdot y = (x_1 y_1, x_2 y_2, \dots)$$

for any $x \in \ell^p$ and $y \in \ell^q$ then $z \in \ell^1$

$$\|z\|_1 \leq \|x\|_p + \|y\|_q$$

Proof. Define $x^{(d)} = (x_1, \dots, x_d)$ then

$$\|z^{(d)}\| \leq \|x^{(d)}\|_p \|y^{(d)}\|_q \leq \|x\|_p \|y\|_q$$

take $d \rightarrow \infty$ we have

$$\|z\|_1 \leq \|x\|_p + \|y\|_q$$

$\square$

Theorem 2.17. (Minkowski inequality) Let $p \in [1, \infty]$, $x, y \in \ell^p \implies (x + y) \in \ell^p$ or equivalently

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p$$

Example. $\ell^\infty = \{x = \{x_n\} : \sup |x_n| < \infty\}$, $c \subset \ell^\infty$ subspace of convergent sequence, $c_0 \subset \ell^\infty$ subspace of $\sup \rightarrow 0$, $c_{00} \subset \ell^\infty$ subspace of $\{x_n\}$ for which there exists a number N such that $x_n = 0$ for all $n \geq N$.

	Banach	Separable
$\ell^p, p < \infty$	+	+
ℓ^∞	+	-
c	+	+
c_0	+	-
$c_{00}, \ \cdot\ _p$	-	+

Proposition 2.18. ℓ^∞ is not separable.

Proof. $\mathcal{B} \subset \ell^\infty$, $x \in \mathcal{B} \Leftrightarrow x_n = 0$ or 1 . For $x, y \in \mathcal{B}$, $\|x - y\|_\infty = 1$. $\mathcal{B}$ is not countable. $B(x, \frac{1}{2})$ disjoint for $x \in \mathcal{B}$. For any dense set, $B(x, \frac{1}{2})$ has to contain an element of it. Then dense set has to contain at least $|\mathcal{B}| = |\mathbb{R}|$ which is uncountable. $\square$

Definition 2.19. (Continuous functions) $C[a, b]$ continuous functions on $[a, b]$ and

$$\|f\|_\infty = \sup_{x \in [a, b]} |f(x)| < \infty$$

and $(C[a, b], \|\cdot\|_\infty)$ is a normed space.

Proposition 2.20. $C[a, b]$ is Banach and separable.

$$f_n \rightarrow f \Leftrightarrow \sup_{x \in [a, b]} |f(x) - f_n(x)| \rightarrow 0$$

f_n uniformly converges to f .

Proof. $\{f_n\}$ is Cauchy sequence of functions, $\forall \varepsilon, \exists N$ such that for all $n, m \geq N$,

$$\sup |f_n(x) - f_m(x)| < \varepsilon$$

Fix x then $\{f_n(x)\}$ is Cauchy and $f(x) := \lim_{n \rightarrow \infty} f_n(x)$. Check $f \in C[a, b]$ and f_n uniformly converges to f . From Analysis 4, uniform convergence limit of continuous function is continuous. $\forall x, |f_n(x) - f_m(x)| < \varepsilon$. Take the limit as $m \rightarrow \infty$ then

$$\sup |f_m(x) - f(x)| < \varepsilon$$

thus it's uniform convergence. Polynomials with coefficient $\mathbb{Q}$ are countable and dense in $C[a, b]$. $\square$

X -complete and some property of elements of x is called generic if it is satisfied on the complement of countable union of nowhere dense sets (meagre set).

Theorem 2.21. Generic element of $C[a, b]$ is nowhere differentiable.

Proof. Define $E_n = \{f \in C[0, 1] : \exists x \in [0, 1 - \frac{1}{n}] \text{ such that } |f(x) - f(x+h)| < nh, \forall h \in (0, 1-x)\}$. We need to do the following steps:

- (1) If f is differentiable at a point then $f \in E_n$ for some n . f is differentiable at $x \in (0, 1)$. $\exists \delta > 0$ such that if $0 < h < \delta$ then

$$|f(x+h) - f(x)| < (|f'(x)| + \varepsilon)h$$

by definition of derivative. For $h > \delta$:

$$|f(x+h) - f(x)| \leq 2\|f\|_\infty \leq \frac{2\|f\|_\infty}{\delta} h$$

$\Rightarrow f \in E_n$ for large n .

- (2a) E_n is closed. Given $\{f_m\} \subset E_n$ where f_m converges uniformly to f . $\exists x_n \in [1 - \frac{1}{n}]$ such that

$$|f_m(x_n) - f_m(x_n+h)| < nh$$

$\forall h \in (0, 1 - x_n)$. Up to the choice of subsequence, assume $x_m \rightarrow x \in (0, 1 - \frac{1}{n})$.

$$\begin{aligned} |f(x+h) - f(x)| &\leq |f(x+h) - f(x_m+h)| + |f(x_m+h) - f_m(x_m+h)| \\ &\quad + |f_m(x_m+h) - f_m(x_m)| + |f_m(x_m) - f(x_m)| + |f(x_m) - f(x)| \\ &< nh + 4\varepsilon \end{aligned}$$

for large m . Thus, $f \in E_n$.

- (2b) E_n^c is dense. $f \in E_n \Rightarrow \exists g$ such that $\|f - g\|_\infty < \varepsilon$ but $g \notin E_n$. $\|\cdot\|_\infty$ is independent of derivatives. Can approximate f by a piecewise linear function $\tilde{f}$ such that $\|f - \tilde{f}\|_\infty < \varepsilon$. This can be shown by proving that f is uniformly continuous. Let M -largest slope of $\tilde{f}$ where $|\tilde{f}'(x)| < M \forall x$. Define $\varphi(x) = \text{dist}(x, \mathbb{Z})$. Define $\tilde{g} = \varepsilon \varphi(\frac{h+M}{\varepsilon}x)$ then $|\tilde{g}'| > n + M$. Let $g = \tilde{g} + \tilde{f}$ then $|g'(x)| > n \Rightarrow g \notin E_n$. At the same time, $\|f - g\| \leq \|f - \tilde{f}\| + \|\tilde{g}\| < 2\varepsilon$. Note that all derivatives here are actually right derivatives.

- (3) E_n is nowhere dense. $\square$

Example. $L^p[a, b]$ with the norm

$$\|f\|_p = \left(\int |f|^p dx \right)^{\frac{1}{p}}$$

$$C_p[a, b] := (C[a, b], \|\cdot\|_p)$$

Proposition 2.22. For $1 \leq p < \infty$. $C_p[a, b]$ is not complete.

Proof. We prove $C_p[a, b] = \bigcup E_n$ where $\{E_n\}$ are nowhere dense then by (BCT), $C_p[a, b]$ is not complete. Define

$$E_n := \{f \in C[a, b] : \|f\|_\infty \leq n\}$$

then $E_n = nE_1$, it suffices to show E_1 is nowhere dense. We will do the following steps:

- (1) E_1 closed. $f_n \rightarrow f$ in $\|\cdot\|_p$ normed then if $|f_n| \leq 1$ then $|f| \leq 1$. Suppose WLOG $\{f > 1 + \varepsilon\}$ for some $\varepsilon > 0$

then

$$\|f - f_n\|_p^p = \int_{\Omega} |f - f_n|^p dx \geq \int_{\Omega} \sup |f - f_n|^p dx \geq \int_{\Omega} \varepsilon^p dx \geq \varepsilon^p(b-a)$$

which is a contradiction.

- (2) Need to show E_1^c dense. Take $f \in E_1 \implies \forall \varepsilon > 0, \exists g \notin E_1$ such that $\|f - g\|_p < \varepsilon$. Let $\varphi(a) = 3, \varphi(x) = 0$ for $x \geq a + \varepsilon$. For any $f \in E_1$, set $g : f + \varphi$ then $g(0) \geq 2$ so $g \notin E_1$ then

$$\|g - f\|_p = \|\varphi\|_p = \left(\int_a^{a+\varepsilon} |\varphi|^p dt \right)^{1/p} \leq 3\varepsilon^{1/p} \rightarrow 0$$

thus, arbitrarily close to any point of E_1 , there exists a point in E_1^c . □

2.2 Precompactness Criteria

Lemma 2.23. X -metric space. Suppose $\forall \varepsilon, \exists \delta, \exists W$ -metric space with $\phi : X \rightarrow W$ such that

- (a) $\phi(X)$ is totally bounded.
- (b) If $\rho_W(\phi(x), \phi(y)) < \delta$ then $\rho_X(x, y) < \varepsilon$.

then X is totally bounded.

Proof. If $\phi(X)$ is totally bounded $\forall \delta > 0, \exists w_i$ with $i \in [1, n]$ such that

$$\begin{aligned} \phi(X) &\subset \bigcup B_W(w_i, \delta) \\ \implies \phi^{-1}(B_W(w_i, \frac{\delta}{2})) &\subset B(z_i, \varepsilon) \end{aligned}$$

where $\phi(z) \in B_W(w_i, \frac{\delta}{2})$. Thus, $X \subset \bigcup B(z_i, \varepsilon)$. □

Proposition 2.24. (precompactness in $\ell^p, p < \infty$) $x \in \ell^p$ and $x^{(n)} = (x_1, \dots, x_n, 0, \dots)$. $A \subset \ell^p$. A is precompact if and only if

- (a) A is bounded.
- (b) $\forall \varepsilon > 0, \exists n > 0$ such that $\forall x \in A$,

$$\|x - x^{(n)}\|_p < \varepsilon$$

Proof. ($\Leftarrow$) Since ℓ^p complete, $A \subset \ell^p$ precompact $\Leftrightarrow A$ is totally bounded. $\varepsilon > 0 \implies n > 0$ such that

$$\|x - x^{(n)}\|_p < \varepsilon, \quad \forall x \in A$$

Take $W = \mathbb{R}^n$ with $\|\cdot\|_p$ norm and $\phi(x) = (x_1, \dots, x_n)$. For $x \in A$, $\|\phi(x)\|_p \leq \|x\|_p$ so $\phi(A)$ is bounded since A is bounded and so totally bounded since $\mathbb{R}^n$ is finitely dimensional. Take $x, y \in A$,

$$\|x - y\|_p \leq \|x - x^{(n)}\|_p + \|x^{(n)} - y^{(n)}\|_p + \|y - y^{(n)}\|_p < 2\varepsilon + \|\phi(x) - \phi(y)\|_p$$

($\Rightarrow$) Totally bounded always implies bounded, so it is sufficient to check condition (2). Suppose that this condition is not satisfied, then for some $\varepsilon > 0$ there exists a sequence $\{x_{(n)}\} \subset A \subset \ell^p$ such that $\|x_{(n)} - x_{(n)}^{(n)}\|_p \geq \varepsilon$. Since A is precompact, there exists $x \in \ell^p$ and a subsequence such that $x_{(n_k)} \rightarrow x$. Since $p < \infty$ one has $\|x - x^{(n)}\|_p \rightarrow 0$ as $n \rightarrow \infty$, hence,

$$\begin{aligned} \|x - x_{(n_k)}\|_p &\geq \|(x_{(n_k)} - x_{(n_k)}^{(n_k)}) - (x_{(n_k)}^{(n_k)} - x_{(n_k)}^{(n_k)}) - (x - x_{(n_k)})\|_p \\ &\geq \|x_{(n_k)} - x_{(n_k)}^{(n_k)}\|_p - \|x_{(n_k)}^{(n_k)} - x_{(n_k)}^{(n_k)}\|_p - \|x - x_{(n_k)}\|_p \geq \frac{\varepsilon}{2} \end{aligned}$$

where the last inequality holds for large enough k . Since $\varepsilon > 0$ is fixed, we obtained a contradiction $\|x - x_{(n_k)}\|_p \rightarrow 0$. □

Theorem 2.25. (precompactness in $C[a, b]$)(Arzela-Ascoli theorem). Let $A \subset C[a, b]$ and A is precompact if and only if

- (a) A is bounded.
- (b) $\forall \varepsilon > 0, \exists \delta > 0$ such that $\forall f \in A$,

$$|f(x) - f(y)| < \varepsilon$$

as long as $|x - y| < \delta$.

Proof. $\forall \varepsilon > 0 \implies \exists \delta > 0$ take partition of $[a, b]$ with mesh $< \delta$ i.e., $a = x_0 < \dots < x_N = b$ with $|x_{i+1} - x_i| < \delta$. Take $W = \mathbb{R}^{N+1}, \|\cdot\|_{\infty}$ and $\phi : C[a, b] \rightarrow W$ where

$$\phi(f) = (f(x_0), f(x_1), \dots, f(x_N))$$

$\forall f \in A, |f(x_i)| \leq \|f\| < C$ so (a) is satisfied. Take $f, g \in A$ take $x_i < x < x_{i+1}$ then

$$|f(x) - g(x)| \leq |f(x) - f(x_i)| + |f(x_i) - g(x_i)| + |g(x_i) - g(x)| \leq 2\varepsilon + \rho_W(\phi(f), \phi(g))$$

then A is totally bounded. □

3 Dual spaces and spaces of operators

Definition 3.1. The operator A is said to be linear if $A : X \rightarrow Y$ then

$$A(ax + by) = aA(x) + bA(y)$$

Definition 3.2. X, Y -normed spaces, then

$$\mathcal{L}(X, Y) = \{A : X \rightarrow Y, A \text{ is linear and continuous}\}$$

Proposition 3.3. Linear operator A -continuous if and only if A is continuous at some point. i.e., $X_n \rightarrow X \implies A(X_n) \rightarrow A(X)$.

Theorem 3.4. Linear operator $A : X \rightarrow Y$ is continuous at 0 if and only if $\exists C > 0$ such that

$$\|A(x)\|_Y \leq C\|x\|_X \quad \forall x \in X$$

Proof. ($\implies$) Continuous at 0 means if $\|x\|_X < \delta \implies \|A(x)\|_Y < 1$ and

$$A(y) = A\left(\frac{y\delta}{\|y\|_X}\right) \frac{\|y\|_X}{\delta}$$

$$\|A(y)\|_Y = \|A\left(\frac{y\delta}{\|y\|_X}\right) \frac{\|y\|_X}{\delta}\|_Y \leq \frac{\|y\|_X}{\delta}$$

($\impliedby$) Let $\varepsilon > 0$, and put $\delta = \varepsilon/C$ then if $x \in B(0, \delta)$ we have

$$\|Ax\|_Y \leq C\|x\|_X \leq \frac{C\varepsilon}{C} = \varepsilon$$

so that A is continuous at 0. □

Definition 3.5. Define

$$\|A\|_{\mathcal{L}(X, Y)} := \sup_{x \neq 0} \frac{\|A(x)\|_Y}{\|x\|_X}$$

Definition 3.6. $\mathcal{L}(X, Y) = \{T : X \rightarrow Y : T \text{ - linear and continuous}\}$ defines a vector space. T sends bounded sets to bounded sets if and only if it is continuous. For that reason, T is often called bounded.

Definition 3.7. $\varphi : X \rightarrow \mathbb{F}$ is continuous operator or bounded functionals then X^* is dual space of X and X^* is always Banach.

Proposition 3.8. $(\mathcal{L}(X, Y), \|\cdot\|_{\mathcal{L}(X, Y)})$ is a normed vector space. If Y is Banach then $\mathcal{L}(X, Y)$ is Banach.

Proof. non-degeneracy and homogeneity are obvious. It suffices to show the triangle inequality

$$\sup_{x \in X \setminus \{0\}} \frac{\|(T_1 + T_2)x\|_Y}{\|x\|_X} \leq \sup_{x \in X \setminus \{0\}} \frac{\|T_1\|_Y + \|T_2\|_Y}{\|x\|_X} \leq \sup_{x \in X \setminus \{0\}} \frac{\|T_1\|_Y}{\|x\|_X} + \sup_{x \in X \setminus \{0\}} \frac{\|T_2\|_Y}{\|x\|_X}$$

thus the triangle inequality holds, i.e., $\|T_1 + T_2\| \leq \|T_1\| + \|T_2\|$. Given $\{T_n\} \subset \mathcal{L}(X, Y)$ we want to show if

$$\sum_{n=1}^{\infty} \|T_n\| < \infty \implies \sum_{n=1}^{\infty} T_n < \infty$$

i.e., $\exists T$ such that

$$\lim_{N \rightarrow \infty} \|T - \sum_{n=1}^N T_n\|_{\mathcal{L}(X, Y)} = 0$$

First to construct, take $x \in X$

$$T(x) = \sum_{n=1}^{\infty} T_n(x)$$

where the terms on the RHS are all in Y so they converge if

$$\sum_{n=1}^{\infty} \|T_n(x)\|_Y < \infty$$

by definition,

$$\sum_{n=1}^{\infty} \|T_n(x)\|_Y \leq \sum_{n=1}^{\infty} \|T_n(x)\|_{\mathcal{L}(X, Y)} \|x\|_X \leq \|x\| \sum_{n=1}^{\infty} \|T_n(x)\|_{\mathcal{L}(X, Y)} < \infty$$

thus,

$$\begin{aligned} T(x) &:= \sum_{n=1}^{\infty} T_n(x) \\ \|T(x)\|_Y &= \lim_{N \rightarrow \infty} \left\| \sum_{n=1}^N T_n(x) \right\| \leq \|x\| \sum_{n=1}^{\infty} \|T_n\|_{\mathcal{L}(X,Y)} \\ \implies \|T\|_{\mathcal{L}(X,Y)} &\leq \sum_{n=1}^{\infty} \|T_n\|_{\mathcal{L}(X,Y)} \end{aligned}$$

apply this to $G_N = T - \sum_{n=1}^N T_n$ then

$$\|G_N\| \leq \sum_{n=N+1}^{\infty} \|T_n\| \rightarrow 0$$

□

Example. $X = C[-1, 1]$, $\|\cdot\|_{\infty}$ -norm and $\psi(f) = f(0)$.

$$\begin{aligned} |\psi(f)| &= |f(0)| \leq \sup_{x \in [-1, 1]} |f(x)| \implies \|\psi\| \leq 1 \\ \implies \|\psi\| &= 1 \end{aligned}$$

$X = C[-1, 1]$, $\|\cdot\|_1$ -norm and $\psi(f) = f(0)$. We want to show that $\exists f_n \in C[-1, 1]$ such that

$$\frac{|\psi(f_n)|}{\|f_n\|} > M$$

More precisely, let's construct f_n such that $\psi(f_n) = 1$ but $\|f_n\|_1 \rightarrow 0$.

$$\int_{-\frac{1}{n}}^{\frac{1}{n}} 1 \, dx = \frac{2}{n} \rightarrow 0$$

Definition 3.9. (Compact operator) $T : X \rightarrow Y$ such that $T(B_X(0, 1))$ is precompact.

Theorem 3.10. (Riesz ℓ^p -representation theorem) Let $p \in [1, \infty)$ and q be the parameter conjugate to p ($p^{-1} + q^{-1} = 1$). Then every bounded linear functional φ on ℓ^p can be uniquely represented as

$$\varphi(x) = \sum_{k=1}^{\infty} x_k y_k$$

for some $y \in \ell^q$. Moreover, $\|\varphi\| = \|y\|_q$. In other words, there is an isomorphism between ℓ^q and $(\ell^p)^*$.

Proof. For $y \in \ell^q$, let $\Psi(y)$ be the linear functional defined by

$$\Psi(y)(x) = \sum_{k=1}^{\infty} x_k y_k$$

we shall show that it is bounded, or i.e. $\Psi_y \in (\ell^p)^*$. By Hölder's inequality, we have

$$|\Psi(y)(x)| \leq \sum_{k=1}^{\infty} |x_k y_k| \leq \|x\|_p \|y\|_q$$

So $\Psi(y)$ is bounded and $\|\Psi(y)\| \leq \|y\|_q$. To see that $\|\Psi(y)\| = \|y\|_q$, define the sequence x as

$$x_k = \|y_k\|^{q-1} \frac{\overline{y_k}}{|y_k|}$$

when $y_k \neq 0$ and $x_k = 0$ when $y_k = 0$. Since q and p are conjugate parameters, $(q-1)p = q$ so that $|x_k|^p = |y_k|^q$, $\|x\|_p^p = \|y\|_q^q$ and $x \in \ell^p$. Furthermore,

$$\begin{aligned} \Psi(y)(x) &= \sum_{k=1}^{\infty} |y_k|^{q-2} y_k \overline{y_k} \\ &= \|y\|_q^q = \|y\|_q \|y\|_q^{q-1} \\ &= \|y\|_q \|x\|_p \end{aligned}$$

so that $\|\Psi(y)\| = \|y\|_q$. This means that Ψ is an isometry and therefore injective. To finish the proof, we need to show that it is surjective. Define the sequence $\{e^{(n)}\} \subset c_{00}$ as

$$e_k^{(n)} = \begin{cases} 1 & n = k \\ 0 & \text{otherwise} \end{cases}$$

Then, every $x \in \ell^p$ can be written as

$$x = \sum_{k=1}^{\infty} x_k e^{(k)}$$

Indeed, let

$$x^{(n)} = \sum_{k=1}^n x_k e^{(k)}$$

and observe that

$$\|x - x^{(n)}\|_p^p = \sum_{k=n+1}^{\infty} |x_k|^p \rightarrow 0$$

as it is the tail of a converging sum. For $\varphi \in (\ell^p)^*$, define y as $y_k = \varphi(e^{(k)})$. We can see that if $y \in \ell^q$, then by linearity $\Psi(y) = \varphi$. Define $\varphi^{(n)} = (y_1, \dots, y_n, 0, \dots)$, and as before define $x^{(n)}$ as

$$x_k^{(n)} = |y_k^{(n)}|^{q-1} \frac{\overline{y_k^{(n)}}}{|y_k^{(n)}|}$$

, so that $\|x^{(n)}\|_p^p = \|y^{(n)}\|_q^q$. We have that

$$\|y^{(n)}\|_q \|x^{(n)}\|_p = \|y^{(n)}\|_q^q = \sum_{k=1}^n |y_k|^q = |\varphi(x^{(n)})| \leq \|\varphi\| \|x^{(n)}\|_p$$

Suppose that $y \notin \ell^q$, then we deduce that $\|\varphi\| = +\infty$, i.e. φ is not bounded, a contradiction. $\square$

Theorem 3.11. (L^p -representation theorem) Let $p < \infty$, $\forall \varphi \in (L^p[a, b])^* \exists g \in L^q[a, b]$ such that $\forall f \in L^p$

$$\varphi(f) = \int_a^b f g \, dx$$

and $\|\varphi\| = \|g\|_q$.

4 Hilbert Spaces

4.1 Definition and orthogonal decomposition

Definition 4.1. X is a vector space over $\mathbb{F}$ take the Hermitian product on X is $(\cdot, \cdot) \rightarrow \mathbb{R}$ satisfies:

(1) (Linear)

$$(x + y, z) = (x, z) + (y, z)$$

(2) (Homogeneous)

$$(\alpha x, y) = \alpha(x, y)$$

(3) (Symmetry)

$$(x, y) = \overline{(y, x)}$$

(4) (Positive definite)

$$(x, x) \geq 0 \quad \text{equality iff } x = 0$$

and $(x, \beta y) = \overline{\beta}(x, y)$ which is called pre-Hilbert space.

Definition 4.2. The norm of X is defined by $\|x\| = \sqrt{(x, x)}$.

Lemma 4.3. (Cauchy-Schwarz inequality)

$$|(x, y)| \leq \|x\| \|y\|$$

Proof. Assume $y \neq 0$ and note that $\|x - \lambda y\|^2 \geq 0$ then

$$(x, x) - \lambda(y, x) - \overline{\lambda}(x, y) + |\lambda|^2 \|y\|^2$$

take

$$\lambda = \frac{(x, y)}{\|y\|^2}$$

it yields

$$\begin{aligned} \|x\|^2 - \frac{(x, y)(y, x)}{\|y\|^2} - \frac{\overline{(x, y)}}{\|y\|^2} (x, y) + \frac{|(x, y)|^2}{\|y\|^2} \|y\|^2 &= \|x\|^2 - \frac{|(x, y)|^2}{\|y\|^2} \geq 0 \\ \implies |(x, y)| &\leq \|x\| \|y\| \end{aligned}$$

□

Corollary 4.4.

$$\|x + y\| \leq \|x\| + \|y\|$$

Proof.

$$\begin{aligned} (x + y, x + y) &= \|x\|^2 + \|y\|^2 + 2 \operatorname{Re}(x, y) \\ &\leq \|x\|^2 + 2|(x, y)| + \|y\|^2 \\ &\leq \|x\|^2 + 2\|x\| \|y\| + \|y\|^2 \\ &= (\|x\| + \|y\|)^2 \end{aligned}$$

□

Definition 4.5. A pre-Hilbert space is called Hilbert space if it is complete with respect to $\|\cdot\|$.

Proposition 4.6. (Parallelogram identity) X -pre-Hilbert space, then

$$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2)$$

Proof.

$$\begin{aligned} \|x + y\|^2 + \|x - y\|^2 &= \|x\|^2 + \operatorname{Re}(x, y) + \|y\|^2 + \|x\|^2 - \operatorname{Re}(x, y) + \|y\|^2 \\ &= 2\|x\|^2 + 2\|y\|^2 \end{aligned}$$

□

Corollary 4.7. ℓ^p are not Hilbert for $p \neq 2$.

Proof. Define

$$\begin{aligned} x &= e^{(1)} = (1, 0, \dots) \\ y &= e^{(2)} = (0, 1, \dots) \end{aligned}$$

then

$$\begin{aligned}\|x + y\|_p^2 &= 2^{2/p} \\ \|x - y\|_p^2 &= 2^{2/p}\end{aligned}$$

and

$$\|x\|_p = \|y\|_p = 1$$

then by Parallelogram identity

$$2^{2/p} + 2^{2/p} = 2(1 + 1)$$

which means $p = 2$. □

Definition 4.8. x, y are orthogonal $x \perp y$ iff $(x, y) = 0$. If A, B are two subspaces then $A \perp B$ iff $\forall x \in A, y \in B$ then $(x, y) = 0$. Orthogonal complement of A -subspace is

$$A^\perp = \{x \in X : x \perp A\}$$

Proposition 4.9. $A^\perp$ is closed subspace and

$$A^\perp \cap A = \{0\}$$

Proof. $x \in A^\perp, y \in A^\perp$ and $z \in A$ then

$$\begin{aligned}(x + y, z) &= (x, z) + (y, z) = 0 \\ (\lambda x, z) &= \lambda(x, z) = 0\end{aligned}$$

Consider $x_n \in A^\perp$ and $x_n \rightarrow x$ then $\forall y \in A$.

$$0 = (x_n, y) \rightarrow (x, y) = 0$$

$$|(x, y) - (x_n, y)| = |(x - x_n, y)| \leq \|y\| \|x - x_n\|$$

thus, $\forall z_n \rightarrow z$ in X we have $(z_n, y) \rightarrow (z, y) \forall y \in X$. □

In linear algebra $H = A \oplus A^\perp$ which is not always true here.

Lemma 4.10. $A \subset H$ subspace and $b \in H \setminus A$. For $a \in A$, the followings are equivalent:

- (1) $a - b \perp A$.
- (2) $\|a - b\| = \inf_{\tilde{a} \in A} \|\tilde{a} - b\|$.

Proof. ((1) $\implies$ (2)) Note

$$\begin{aligned}\|\tilde{a} - b\|^2 &= (\tilde{a} - b, \tilde{a} - b) \\ &= (\tilde{a} - a + a - b, \tilde{a} - a + a - b) \\ &= \|\tilde{a} - a\|^2 + \|a - b\|^2 + 2\operatorname{Re}(\tilde{a} - a, a - b) \\ &= \|\tilde{a} - a\|^2 + \|a - b\|^2 \geq \|a - b\|^2\end{aligned}$$

((2) $\implies$ (1)) Take $\tilde{a} \in A$ then define

$$f_t(a) := \|a - b - t\tilde{a}\|^2$$

this has minimum at $t = 0$. Expanding the function we have

$$\begin{aligned}f_t(a) &= \|a - b\|^2 + t^2\|\tilde{a}\|^2 + 2t\operatorname{Re}(a - b, \tilde{a}) \\ \implies \frac{d}{dt}f_t(a)|_{t=0} &\implies \operatorname{Re}(a - b, \tilde{a}) = 0 \quad \forall \tilde{a} \in A\end{aligned}$$

□

Theorem 4.11. Let $A \subset H$ be closed subspace. Then $H = A \oplus A^\perp$.

Proof. Fix $x \in H$, $\inf_{a \in A} \|x - a\| = \delta$ is achieved for some a . i.e., $\exists$ sequence (a_n) such that $\|x - a_n\| \rightarrow \delta$. It is enough to show it's Cauchy then it converges. Take n, m large then

$$\begin{aligned}\|a_n - a_m\|^2 &= 2\|x - a_n\|^2 + 2\|x - a_m\|^2 - 4\|x - \frac{a_n + a_m}{2}\|^2 \\ &\leq 2\delta^2 + 2\delta^2 - 4\delta^2 < \varepsilon\end{aligned}$$

then (a_n) is Cauchy which means $x - a \perp A$ and $x = a + (x - a)$. Now to show the uniqueness, let $x = a + b = a' + b'$ where $a, a' \in A$ and $b, b' \in A^\perp$ then

$$A \ni a - a' = b - b' \in A^\perp$$

which is possible if and only if $a - a' = b - b' = 0$ so the uniqueness is achieved. □

Definition 4.12. If $H = A \oplus A^\perp$ and A -closed then $P_A : H \rightarrow H$ is orthogonal projection onto A where $P_A(x) = a$ if $x = a + b$ where $a \in A$ and $b \in A^\perp$.

Proposition 4.13.

- (i) P_A is linear.
- (ii) $\|P_A(x)\| \leq \|x\|$.
- (iii) $P_A^2 = P_A \circ P_A = P_A$.
- (iv) $\ker(P_A) = A^\perp$.
- (v) $\|P_A\| = 1$ as long as $A \neq \{0\}$.

Proof.

- (i) $x + \alpha y - P_A(x + \alpha y) \perp A$ and $x - P_A(x) \perp A$, $\alpha y - \alpha P_A(y) \perp A$ then

$$P_A(x + \alpha y) - P_A(x) - \alpha P_A(y) \perp A$$

then $P_A(x + \alpha y) = P_A(x) + \alpha P_A(y)$ so it's linear.

- (ii) $x = P_A(x) + (x - P_A(x))$ so

$$\|x\|^2 = \|P_A(x)\|^2 + \|x - P_A(x)\|^2 \geq \|P_A(x)\|^2$$

- (iii) $P_A(P_A(x)) = P_A(a) = a = P_A(x)$.

- (iv) $x = P_A(x) + (x - P_A(x))$

$$P_A(x) = 0 \Leftrightarrow x = (x - P_A(x)) \in A^\perp$$

□

Example. $A = \mathbb{F}e$ is one-dimensional subspace in H and $e = H \setminus \{0\}$. A is closed because finite dimensional subspace is closed. Then

$$\begin{aligned} x - P_A(x) \perp A &\implies x - P_A(x) \perp e \\ \implies (x - P_A(x), e) = 0 &\implies (x - ce, e) = 0 \end{aligned}$$

therefore,

$$c = \frac{(x, e)}{(e, e)}$$

4.2 Continuous Linear Functionals

H -Hilbert then $\varphi_y(x) = (x, y)$ for $y \neq x$ is a linear functional then

$$\begin{aligned} \|\varphi_y(x)\| &= |(x, y)| \leq \|x\| \|y\| \\ \implies \|\varphi_y\| &\leq \|y\| \end{aligned}$$

and $\varphi_y(y) = \|y\|^2 = \|y\| \cdot \|y\| \implies \|\varphi_y\| = \|y\|$. Now define

$$\begin{aligned} I : y &\mapsto \varphi_y \\ H &\rightarrow H^* \end{aligned}$$

Proposition 4.14. I is norm preserving and antilinear and

$$I(\alpha y) = \bar{\alpha} I(y)$$

Proposition 4.15. $\forall A \in H$,

- (1) $A^\perp = (\bar{A})^\perp$, i.e. A is dense if and only if $A^\perp = \{0\}$.
- (2) $(A^\perp)^\perp = \bar{A}$.

Proof.

- (1) Let $A \ni x_n \rightarrow x \in \bar{A}$ if

$$(y, x_n) = 0 \implies (y, x) = 0 \implies A^\perp \subset (\bar{A})^\perp$$

And we know that $A \subset \bar{A}$ so $A^\perp \supset (\bar{A})^\perp$ then $A^\perp = (\bar{A})^\perp$.

- (ii) On one hand

$$H = \bar{A} \oplus (\bar{A})^\perp = \bar{A} \oplus A^\perp$$

and similarly

$$\begin{aligned} H &= A^\perp \oplus (A^\perp)^\perp \\ \implies \bar{A} &= (A^\perp)^\perp \end{aligned}$$

□

Theorem 4.16. (Riesz-Fréchet representation theorem) $\forall \varphi \in H^*$, $\exists! y \in H$ such that $\varphi = I(y)$.

Proof. Suppose $\varphi \neq 0$ which means $\ker(\varphi) \neq H$ and closed. Then $\ker(\varphi)^\perp \neq \{0\}$. On one hand,

$$x = \left(x - \frac{\varphi(x)}{\varphi(e)}e\right) + \frac{\varphi(x)}{\varphi(e)}e$$

where

$$x - \frac{\varphi(x)}{\varphi(e)}e \in \ker(\varphi), \quad \frac{\varphi(x)}{\varphi(e)}e \in \ker(\varphi)^\perp$$

which implies $\ker(\varphi)^\perp$ is one-dimensional. At the same time

$$x = (x - \frac{(x, e)}{(e, e)}e) + \frac{(x, e)}{(e, e)}e$$

where

$$\begin{aligned} \frac{(x, e)}{(e, e)}e &\in \ker(\varphi)^\perp, \quad x - \frac{(x, e)}{(e, e)}e \in (\ker(\varphi)^\perp)^\perp \\ &\implies \frac{\varphi(x)}{\varphi(e)} = \frac{(x, e)}{(e, e)} \\ &\implies \varphi(x) = (x, \frac{e\varphi(e)}{(e, e)}) = (x, y) \end{aligned}$$

□

4.3 Orthonormal bases

Definition 4.17. Let $\mathcal{E} = \{e_j : j \in J\} \subset H$ we say $\mathcal{E}$ is

- (1) Orthogonal if $e_j \perp e_k$ for $j, k \in J$.
- (2) Orthonormal if additionally $\|e_j\| = 1$.
- (3) A Hilbert basis if there is no $x \in H$ such that $\mathcal{E} \cup \{x\}$ is orthonormal.

Example. Let $\mathcal{E} = \{e_1, \dots, e_n\}$ and $M = \text{Span}\mathcal{E}$ then

$$P_M = \sum_{k=1}^n \frac{(x, e_k)}{(e_k, e_k)} e_k$$

Theorem 4.18. (Gram-Schmidt orthonormalisation) Let $\{v_1, \dots, v_n\}$ be a collection of vectors then $\exists \mathcal{E}$ be orthonormal set such that

$$\text{Span}\{v_1, \dots, v_n\} = \text{Span}\mathcal{E}$$

Theorem 4.19. All Hilbert spaces have Hilbert bases.

Theorem 4.20. (Bessel's inequality) $\mathcal{E}$ is orthonormal set $\{e_n, n \in \mathbb{N}\}$. Then $\forall x \in H$

$$\sum_{n=1}^{\infty} |(x, e_n)|^2 \leq \|x\|^2$$

with operator

$$P(x) = \sum_{n=1}^{\infty} (x, e_n) e_n$$

is projection onto $\overline{\text{Span}\mathcal{E}}$.

Remark. Bessel's inequality also works when $\mathcal{E}$ is uncountable.

Theorem 4.21. $\mathcal{E} \subset H$ orthonormal set then the followings are equivalent:

- (1) $\mathcal{E}$ is a Hilbert basis.
- (2) If $x \perp \mathcal{E} \implies x = 0$.
- (3) $\overline{\text{Span}(\mathcal{E})} = H$.
- (4) $x = \sum_{e \in \mathcal{E}} (x, e)e$ for all $x \in H$.
- (5) $(x, y) = \sum_{e \in \mathcal{E}} (x, e)(e, y)$ for all $x, y \in H$.
- (6) (Parseval's identity) $\|x\|^2 = \sum_{e \in \mathcal{E}} |(x, e)|^2$ for all $x \in H$.

Proof. ((1) $\implies$ (2)). If $\mathcal{E}$ is Hilbert, $x \perp \mathcal{E}$. If $x \neq 0$ then $\mathcal{E} \cup \{\frac{x}{\|x\|}\}$ is an orthonormal basis containing $\mathcal{E}$.

((2) $\implies$ (3)). $\text{Span}(\mathcal{E})^\perp = \overline{\text{Span}(\mathcal{E})}^\perp$ then

$$H = \{0\}^\perp = \overline{\text{Span}(\mathcal{E})}$$

((3) $\implies$ (4)). If $\overline{\text{Span}(\mathcal{E})} = H$ then $P_L = Id$ and $x = P_L x = \sum (x, e)e$.

((4) $\implies$ (5)).

$$(x, y) = (\sum_{e \in \mathcal{E}} (x, e)e, y) = \sum_{e \in \mathcal{E}} (x, e)(e, y)$$

((5) $\implies$ (6)). Take $x = y$.
 ((6) $\implies$ (2)).

$$\|x\|^2 = \sum |(x, e)|^2 = 0 \implies x = 0$$

□

Definition 4.22. The series in

$$x = \sum_{e \in \mathcal{E}} (x, e) e$$

is called the Fourier series of x with respect to $\mathcal{E}$. And (x, e) is known as the Fourier coefficient.

Theorem 4.23. Assume that H has a countable Hilbert basis, $\mathcal{E} = \{e_n\}_{n \in \mathbb{N}}$ then $\exists F : H \rightarrow \ell^2$ such that

(a) F is linear

$$F(ax + by) = aF(x) + bF(y)$$

(b) F preserves inner product

$$(Fx, Fy)_{\ell^2} = (x, y)_H$$

(c) F is a bijection.

Proof. $F(x)_k = (x, e_k)$ then (a) is obvious. Then to check (b)

$$(F(x), F(y))_{\ell^2} = \sum (x, e_k) \overline{(y, e_k)} = (x, y)$$

which implies F is distance-preserving so F is injective. It suffices to show the surjectivity. Let $\xi = \{\xi_k\} \in \ell^2$ i.e. $\sum |\xi_k|^2 < +\infty$. Want to show $\exists x \in H$ such that $F(x) = \xi$ then

$$x = \sum_{k=1}^{\infty} \xi_k e_k$$

to show the convergence

$$\begin{aligned} \left\| \sum_{k=n}^m \xi_k e_k \right\|^2 &= \left(\sum_{k=n}^m \xi_k e_k, \sum_{k=n}^m \xi_k e_k \right) \\ &= \sum_{k=n}^m |\xi_k|^2 < \varepsilon \end{aligned}$$

for large n, m since $\sum |\xi_k|^2 < \infty$.

□

Theorem 4.24. H has countable basis if and only if H is separable.

Proof. ($\Leftarrow$) Apply Gram-Schmidt orthonormalisation to the dense set.

□

Example. Consider $L^2(-\pi, \pi)$ and $\mathcal{E} = \left\{ \frac{1}{\sqrt{2\pi}} e^{ikt} \right\}_{k \in \mathbb{Z}}$. Then

$$\begin{aligned} (e_k, e_l) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ikt} \overline{e^{ilt}} dt \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(k-l)t} dt = \delta_{kl} \end{aligned}$$

Claim: $\mathcal{E}$ is Hilbert basis, i.e. $\text{Span}(\mathcal{E}) = L^2[-\pi, \pi]$.

Exercise. Consider $L^2[1/2, 2]$, given that $M = \{f \in L^2[1/2, 2] : f(x) = f(1/x)\}$. Show that M is a closed subspace of $L^2[1/2, 2]$ and find $M^\perp$ and P_M .

We are only interested in finding $M^\perp$ and P_M which are the nasty things in this problem. To find $M^\perp$, suppose $f(x) \in M$ and $g(x) \in M^\perp$ then

$$\int_{1/2}^2 f(x)g(x) dx = 0$$

Substitute by $u = 1/x$ yields

$$\int_{1/2}^2 f(1/x)g(1/x)x^2 dx = 0$$

Since $f(x) = f(1/x)$,

$$\begin{aligned} \int_{1/2}^2 f(x)g(1/x)x^2 dx &= 0 \\ \implies \int_{1/2}^2 f(x)(g(x) + g(1/x)x^2) dx &= 0 \end{aligned}$$

since this works for all $f(x) \in M$, a nontrivial solution for g is $g(x) + g(1/x)x^2 = 0$. Now, we are left to prove that any

functions $h(x) \in L^2[1/2, 2]$ can be written as $h(x) = h_1(x) + h_2(x)$ where $h_1(x) \in M$ and $h_2(x) \in M^\perp$. By observation,

$$p(x) = \frac{h(x)}{1+x^2} \implies p(1/x) = \frac{x^2 h(1/x)}{1+x^2}$$

and

$$q(x) = \frac{x^2 h(x)}{1+x^2} \implies q(1/x) = \frac{h(1/x)}{1+x^2}$$

Then it's natural to think about the function

$$h_1(x) = \frac{h(x)}{x^2+1} + \frac{x^2 h(1/x)}{x^2+1}$$

where $h_1(x) = h_1(1/x)$. We can check that

$$h_2(x) = \frac{x^2}{x^2+1} (h(x) - h(1/x))$$

is indeed the function we need

$$h_2(1/x) = \frac{1}{x^2+1} (h(1/x) - h(x)) = -\frac{h_2(x)}{x^2}$$

i.e. $x^2 h_2(1/x) + h_2(x) = 0$ so $h_2(x) \in M^\perp$. Thus,

$$P_M(h(x)) = h_1(x) = \frac{h(x) + x^2 h(1/x)}{x^2+1}$$

4.4 Adjoint operators

Definition 4.25. Let H be a Hilbert space and $\mathcal{L}(H) = \{A : H \rightarrow H : \text{bounded operator}\}$. Let $A \in \mathcal{L}(H)$, A^* is adjoint of A then $A^* \in \mathcal{L}(H)$ with

$$(Ax, y) = (x, A^*y) \quad \forall x, y \in H$$

Proposition 4.26. $\forall A \in \mathcal{L}(H)$, there exists unique $A^* \in \mathcal{L}(H)$.

Proof. Let $y \in H$, define

$$\varphi(x) = (Ax, y)$$

To show $\varphi(x)$ is a linear functional, by Cauchy-Schwartz inequality

$$\begin{aligned} |\varphi(x)| &= |(Ax, y)| \leq \|Ax\| \|y\| \leq \|A\| \|x\| \|y\| \\ \implies \|\varphi\| &\leq \|A\| \|y\| \end{aligned}$$

By Rieze-representation theorem, there exists some $z = A^*y$ such that

$$(Ax, y) = (x, A^*y)$$

We need to check that A^* is linear and bounded. Linearity is trivial so we only need to check the boundedness. In fact, by Rieze-representation theorem so

$$\begin{aligned} \|\varphi\| &= \|z\| = \|A^*y\| \leq \|A\| \|y\| \\ \implies \|A^*\| &\leq \|A\| \end{aligned}$$

Now to show the uniqueness. Suppose A' is another adjoint of A then

$$\begin{aligned} (x, A^*y) &= (Ax, y) = (x, A'y) \quad \forall x, y \\ \implies (x, (A^* - A')y) &= 0 \quad \forall x, y \end{aligned}$$

Take $x = (A^* - A')y$ then we immediately have

$$\|(A^* - A')y\| = 0 \implies A^* = A'$$

□

Proposition 4.27.

1. $(A^*)^* = A$.
2. $(\alpha A + \beta B)^* = \bar{\alpha} A^* + \bar{\beta} B^*$.
3. $(AB)^* = B^* A^*$

Proof.

1. $(Ax, y) = (x, A^*y) = \overline{(A^*y, x)} = \overline{(y, A^{**}x)} = (A^{**}, y), \forall x, y$.
2. $((\alpha A + \beta B)x, y) = \alpha(Ax, y) + \beta(Bx, y) = \alpha(x, A^*y) + \beta(x, B^*y) = (x, (\bar{\alpha} A^* + \bar{\beta} B^*)y)$.
3. $(ABx, y) = (Bx, A^*y) = (x, B^* A^*y)$.

□

Corollary 4.28. $\|A^*\| = \|A\|$.

Proof. We have shown that $\|A^*\| \leq \|A\|$ then use the above proposition yields $\|A^{**}\| \leq \|A^*\| \implies \|A\| \leq \|A^*\|$ then $\|A^*\| = \|A\|$. $\square$

Example. Let H be a Hilbert space and $U \in \mathcal{L}(H)$ preserves the inner product. Then $U^*U = I$. By definition of isometry, $\forall x, y$

$$(x, y) = (Ux, Uy) = (U^*Ux, y)$$

Example. Let $L \subset H$ be a closed subspace then by definition

$$x = P_L x + (1 - P_L)x$$

Claim: $P_L^* = P_L$ i.e.

$$(P_L x, y) = (x, P_L y)$$

Proof. Suppose

$$x = P_L x + (1 - P_L)x$$

$$y = P_L y + (1 - P_L)y$$

Then

$$(P_L x, y) = (P_L x, P_L y) = (x, P_L y)$$

since $P_L \perp (1 - P_L)$. $\square$

5 Hahn-Banach Theorem

Definition 5.1. (Convex sets) $A \subset X$. We say A is convex if $x, y \in A$, $t \in [0, 1]$ then $tx + (1 - t)y \in A$.

Lemma 5.2. Let A, B convex where

$$A + B := \{x = a + b : a \in A, b \in B\}$$

Then $A + B$ is convex.

Proof. Let

$$x = a_x + b_x \in A + B$$

$$y = a_y + b_y \in A + B$$

then

$$\begin{aligned} tx + (1 - t)y &= ta_x + tb_x + (1 - t)a_y + (1 - t)b_y \\ &= (ta_x + (1 - t)a_y) + (tb_x + (1 - t)b_y) \in A + B \end{aligned}$$

□

Corollary 5.3. Let $\mathcal{A}$ be a collection of convex sets then

$$B = \bigcap_{A \in \mathcal{A}} A$$

is convex.

Lemma 5.4. X be a normed space. If A, B is open then $A + B$ is open.

Proof.

$$A + B = \bigcup_{b \in B} \{b\} + A$$

then the translation of an open set is open.

□

Lemma 5.5. A is closed, B is compact then $A + B$ is closed.

Proof. Let $x_n \in A + B$ with $x_n = a_n + b_n$ where $x_n \rightarrow x$ then we want to show $x \in A + B$. Since B is compact so $\exists b_{n_k} \rightarrow b \in B$. Then

$$a_{n_k} \rightarrow x - b \in A$$

since A is closed. Thus, $x = (x - b) + b \in A + B$.

□

Definition 5.6. X -vector space. Then $p : X \rightarrow \mathbb{R}$ is called the convex functional if

1. $p(x) \geq 0$.
2. $p(\alpha x) = \alpha p(x)$, $\forall \alpha \geq 0$.
3. $p(x + y) \leq p(x) + p(y)$

Example. X -normed space over $\mathbb{R}$, then $p(x) = \|x\|$ -convex.

Definition 5.7. X -normed over $\mathbb{R}$ then $C \subset X$ be a convex set, if $0 \in C$, C is open. Define $p : X \rightarrow \mathbb{R}$ be Minkowski functional or convex gauge if

$$p(x) = p_C(x) = \inf_{\alpha > 0} \{\alpha > 0 : \frac{x}{\alpha} \in C\}$$

Lemma 5.8.

- (1) $C = \{x \in X : p(x) < 1\}$.
- (2) $\exists M$, such that $0 \leq p(x) \leq M\|x\|$.
- (3) p is convex functional.

Proof.

- (1) Take $x \in C$ then $p(x) \leq 1$. If $p(x) = 1$, $\exists \alpha_n \uparrow 1$ such that $\frac{x}{\alpha_n} \notin C$ where $\frac{x}{\alpha_n} \rightarrow x$ contradicting that C is open. If $p(x) = \beta$, then $\forall \alpha > \beta$, $\frac{x}{\alpha} \in C$.

(2) Let C be an open set containing 0, i.e. $B(0, \varepsilon) \subset C$ so

$$\begin{aligned} \frac{\varepsilon}{\|x\|}x &\in C \\ \implies p(x) &\leq \frac{\|x\|}{x} \end{aligned}$$

(3) $p(\beta x) = \beta p(x)$ with $\beta > 0$ then

$$\frac{x}{\alpha} \in C \implies \frac{\beta x}{\beta \alpha} \in C$$

Let $\alpha_n \downarrow p(x)$, $\beta_n \downarrow p(y)$ then

$$\begin{aligned} \frac{x}{\alpha_n} \in C, \quad \frac{y}{\beta_n} \in C \\ \frac{\alpha_n}{\alpha_n + \beta_n} \frac{x}{\alpha_n} + \frac{\beta_n}{\alpha_n + \beta_n} \frac{y}{\beta_n} \in C \\ \implies \frac{x+y}{\alpha_n + \beta_n} = p(x+y) \leq \alpha_n + \beta_n = p(x) + p(y) \end{aligned}$$

□

Theorem 5.9. (Hahn-Banach theorem) Let X be a vector space, $p : X \rightarrow \mathbb{R}$ is convex functional and $W \subset X$ is subspace. Let $\varphi : W \rightarrow \mathbb{R}$ be linear functional satisfying $\varphi(w) \leq p(w)$, $\forall w \in W$. Then $\exists \phi : X \rightarrow \mathbb{R}$ such that

1. ϕ is linear.
2. $\phi(w) = \varphi(w)$, $\forall w \in W$.
3. $\phi(x) \leq p(x)$, $\forall x \in X$.

Proof. Suppose $W \neq X$, $\exists y \in X \setminus W$. We want to show $Y = W + \mathbb{R}y$ then holds with $X = \tau_n$ i.e. $\exists$ extension F to Y .

$$F(w + ty) = \varphi(w) + ta$$

where $a = F(y)$. Need this to satisfy:

$$\varphi(w) + ta \leq p(w + ty)$$

For $t > 0$:

$$\varphi\left(\frac{w_1}{t}\right) + a \leq p\left(\frac{w_1}{t} + y\right)$$

For $t < 0$, we divide by $|t|$ we get

$$\varphi\left(\frac{w_2}{|t|}\right) - a \leq p\left(\frac{w_2}{|t|} - y\right)$$

We need a, b satisfy

$$\begin{aligned} \varphi(w_2) - p(w_2 - y) \leq a \leq p(w_1 + y) - \varphi(w_1) \\ \varphi(w_1) + \varphi(w_2) \leq p(w_1 + w_2) \leq p(w_1 + y) + p(w_2 - y) \end{aligned}$$

By convexity. Thus, there exists some a such that

$$\varphi(w_2) - p(w_2 - y) \leq a \leq p(w_1 + y) - \varphi(w_1)$$

□

Theorem 5.10. (a version of Hahn-Banach theorem) Let X be a normed space, $W \subset X$ is a linear subspace and $\varphi \in W^*$. Then there exists an extension of φ denoted by $\psi \in X^*$ such that $\|\psi\|_{X^*} = \|\varphi\|_{W^*}$.

Proof. We know that

$$\sup_{w \neq 0} \frac{|\varphi(w)|}{\|w\|} = \|\varphi\|_{W^*}$$

Thus, $\varphi(w) \leq \|\varphi\|_{W^*} \|w\|$. Define

$$p(x) := \|\varphi\|_{W^*} \|x\| \quad \forall x \in X$$

Then p is homogeneous, subadditive and $\varphi(w) \leq p(w)$, $\forall w \in W$. By Theorem 5.9, we can extend $\psi : X \rightarrow \mathbb{R}$ such that $\psi(x) \leq p(x) = \|\varphi\|_{W^*} \|x\|$, $\forall x \in X$. Then

$$\begin{aligned} \frac{|\psi(x)|}{\|x\|} &\leq \|\varphi\|_{W^*} \\ \implies \sup_{x \neq 0} \frac{|\psi(x)|}{\|x\|} &\leq \|\varphi\|_{W^*} \implies \|\psi\|_{X^*} \leq \|\varphi\|_{W^*} \end{aligned}$$

And by definition of dual norm $\implies \|\psi\|_{X^*} \geq \|\varphi\|_{W^*}$ so this completes the proof.

□

Corollary 5.11. Let $x \in X$, X is normed. Then $\exists \varphi \in X^*$ such that $\|\varphi\| = 1$ and $\varphi(x) = \|x\|$.

Proof. Let $x \neq 0$, put $W = \text{span}\{x\} = \{\lambda x, \lambda \in \mathbb{R}\}$ and define $\varphi(\lambda x) = \lambda \|x\|$. Then $\varphi : W \rightarrow \mathbb{R}$ and

$$\|\varphi\| = \sup_{x \neq 0} \frac{|\varphi(\lambda x)|}{\|\lambda x\|} = \sup_{x \neq 0} \frac{|\lambda| \|x\|}{\|\lambda x\|} = 1$$

extend to X by Hahn-Banach theorem.

□

Corollary 5.12. Let $x \in X$, then

$$\|x\| = \sup_{\varphi \in X^*, \|\varphi\|=1} |\varphi(x)|$$

Definition 5.13. Let X be a topological vector space over $\mathbb{R}$, $\alpha \in \mathbb{R}$, $\varphi \in X^* \setminus \{0\}$. The closed hyperplane of equation $[\varphi = \alpha]$ is the set

$$E := [\varphi = \alpha] = \{x \in X | \varphi(x) = \alpha\}$$

Lemma 5.14. Let $y \in X$ satisfies $\varphi(y) = \alpha$. Then

$$[\varphi = \alpha] = y + \ker(\varphi) = \{y + x | \varphi(x) = 0\}$$

Proof. Just as a result of linearity of φ . □

Definition 5.15. Let $A, B \subset X$, the hyperplane $[\varphi = \alpha]$ separates A and B , if $\varphi(x) \leq \alpha$, $\forall x \in A$ and $\varphi(y) \geq \alpha$, $\forall y \in B$ (or reverse the inequalities). We say that $[\varphi = \alpha]$ strictly separated A and B if given $\varepsilon > 0$

$$\varphi(x) \leq \alpha - \varepsilon < \alpha + \varepsilon \leq \varphi(y), \quad \forall x \in A, y \in B$$

Lemma 5.16. X -normed over $\mathbb{R}$, $C \subset X$ where C is convex and open. Let $x \in X \setminus C$. Then there exists $\varphi \in X^* \setminus \{0\}$ such that $[\varphi = \varphi(x)]$ separates x and $\{C\}$.

Proof. We may assume that $0 \in C$. Indeed, if $[\varphi = \varphi(x)]$ separates C and x , then $C - y$ and $x - y$ are separated by $[\varphi = \varphi(x) - \varphi(y)]$. $\forall c, \varphi(c) \leq \varphi(x) \Leftrightarrow \varphi(c - y) \leq \varphi(x) - \varphi(y)$. So WLOG, $0 \in C \Rightarrow \exists p_C$ -convex functional, s.t. $C = \{x | p_C(x) < 1\}$. Define $\varphi : \mathbb{R}_X \rightarrow \mathbb{R}$ by $\varphi(tx) = t$. Then $p_C(tx) = tp_C(x) \geq t$ if $t > 0$ and $p_C(tx) \geq 0$ if $t \leq 0$ so $\varphi(tx) \leq p_C(tx)$. Thus, by Hahn-Banach, there is an extension of φ to X , s.t.

$$\varphi(y) \leq p_C(y) \quad \forall y \in X$$

In particular, $y \in C$, then

$$\varphi \leq p_C(y) < 1 \Rightarrow \varphi(x) = 1$$

□

Theorem 5.17. (First Hahn-Banach separation theorems) X -normed over $\mathbb{R}$, let A, B be two convex disjoint sets where A is open. Then there exists $\varphi \in X^* \setminus \{0\}$ and $\alpha \in \mathbb{R}$ such that $[\varphi(x) = \alpha]$ separates A and B . Moreover,

$$A \subset \{\varphi(x) < \alpha\}$$

i.e. $\varphi(a) < \alpha, \forall a \in A$.

Proof. To separate A from B is equivalent to separate $A - B$ from $\{0\}$. Indeed if

$$\varphi(a) \geq \alpha, \varphi(b) \leq \alpha, \quad \forall a, b$$

$$\varphi(a) - \varphi(b) \geq 0 = \varphi(0), \quad \forall a, b$$

Similarly, if $c \in A - B$ then

$$\varphi(a) - \varphi(b) = \varphi(c) \geq 0$$

Define $\alpha = \inf_{a \in A} \varphi(a)$ then

$$\varphi(0) \geq \alpha \geq \varphi(b)$$

A -open implies $A - B$ is open so we can separated it from any point outside of $A - B$. In particular, since $A \cap B = \emptyset$, $0 \notin A - B$ so $A - B$ can be separated from $\{0\}$. Suppose $\exists a \in A$ such that $\varphi(a) = \alpha$, $\exists \varepsilon > 0$ such that $B(a, \varepsilon) \subset A$. $\exists x \in X$ such that $\varphi(x) > 0$ and $\|x\| = 1$. Then $a + \frac{\varepsilon}{2}x \in B(a, \varepsilon) \subset A$. On one hand

$$\varphi(a + \frac{\varepsilon}{2}x) \leq \alpha$$

$$\varphi(a + \frac{\varepsilon}{2}x) = \varphi(a) + \frac{\varepsilon}{2}\varphi(x) = \alpha + \frac{\varepsilon}{2}\varphi(x) > \alpha$$

so $A \not\subset [\varphi \leq \alpha]$, a contradiction. □

Remark. Fix $\varphi \in X^* \setminus \{0\} \Rightarrow A = \{\varphi > 0\}, B = \{\varphi < 0\}$.

Theorem 5.18. (Second Hahn-Banach) X -normed over $\mathbb{R}$, A, B are two disjoint convex subsets of A where A is compact and B is closed. Then $\exists \varphi \in X^* \setminus \{0\}$, $\exists \alpha \in \mathbb{R}$ such that $[\varphi = \alpha]$ strictly separates A and B .

Proof. Strictly separating A and B is equivalent to strictly separating $A - B$ and $\{0\}$. $A \cap B = \emptyset \Rightarrow 0 \notin A - B$ and $A - B$ is closed and convex. $\exists \varepsilon > 0$ such that $B(0, \varepsilon) \cap A - B = \emptyset$. Any ball is in fact convex. Then by previous theorem, $\exists \varphi, \alpha$ such that

$$\varphi(x) < \alpha \leq \varphi(y), \quad \forall x \in B(0, \varepsilon), y \in A - B$$

then $0 < \alpha \leq \varphi(a) - \varphi(b)$, $\forall a \in A, b \in B$. Consider $\beta = \inf_{a \in A} \varphi(a)$, $\gamma = \sup_{b \in B} \varphi(b)$ then

$$\varphi(b) < \frac{\alpha}{2} + \varphi(b) < \alpha + \varphi(b) \leq \varphi(a)$$

$$\implies \varphi(b) \leq \gamma < \frac{\alpha}{2} + \gamma < \alpha + \gamma \leq \beta \leq \varphi(a)$$

Hence, $\varphi = [\frac{\alpha}{2} + \gamma]$ strictly separates A and B .

□

6 Applications of BCT to linear operators

In linear algebra, $A : V \rightarrow W$ is linear operator bijective then A^{-1} is linear operator. While in functional analysis, $A : V \rightarrow W$ is bounded linear bijective then A^{-1} is linear, is it bounded?

Definition 6.1. A is nowhere dense if one the followings hold:

- (1) $\overline{A}$ has empty interior.
- (2) $(\overline{A})^c$ is dense.
- (3) $\forall$ open $V \subset X, \exists V \subset U$ such that $V \cap A = \emptyset$.

Theorem 6.2. (Remainder on Baire Category theorem) Complete metric X cannot be decomposed as a countable union of nowhere dense sets.

$$X = \bigcup_{i=1}^{\infty} A_i \implies$$

$\exists i$ such that $\text{int}(\overline{A_i}) \neq \emptyset$.

Theorem 6.3. (Banach-Steinhaus theorem) Let X -Banach and Y -normed. Let

$$\mathcal{T} = \{T_j \in \mathcal{L}(X, Y), j \in J\}$$

Suppose that $\forall x \in X,$

$$\sup_{j \in J} \|T_j(x)\|_Y < +\infty$$

then

$$\sup_{j \in J} \|T_j\| < +\infty$$

i.e. $\exists C > 0$, such that $\forall j \in J, \forall x \in X,$

$$\|T_j(x)\| \leq C\|x\|$$

Proof. Let

$$X_n = \{x \in X : \|T_j(x)\| \leq n, \forall j \in J\}$$

then

$$X = \bigcup_{n=1}^{\infty} X_n$$

By BCT, $\exists m$ such that $\overline{X_m}$ has non-empty interior. Note that

$$X_n = \bigcap_{j \in J} T_j^{-1}(B[0, n])$$

is closed. Thus, X_m has non-empty interior. $\exists x_0 \in X$ and $\varepsilon > 0$, such that $B(x_0, \varepsilon) \subset X_m$. Take $x \in X$ with $\|x\| = 1$ then note

$$x = \frac{2}{\varepsilon}[(x_0 + \frac{\varepsilon x}{2}) - x_0]$$

so

$$\begin{aligned} \|T_j(x)\| &= \frac{2}{\varepsilon} \|T_j(x_0 + \frac{\varepsilon x}{2}) - T_j(x_0)\| \\ &\leq \frac{2}{\varepsilon} (\|T_j(x_0 + \frac{\varepsilon x}{2})\| + \|T_j(x_0)\|) \\ &\leq \frac{2}{\varepsilon} (m + m) = \frac{4m}{\varepsilon} = C \end{aligned}$$

□

Corollary 6.4. X -Banach, Y -normed. Let $\{T_n\} \subset \mathcal{L}(X, Y)$ be sequence of operators such that

$$T_n(x) \rightarrow T(x), \quad x \in X$$

then:

- (1) $\sup_n \|T_n\| < +\infty$.
- (2) $T \in \mathcal{L}(X, Y)$.
- (3) $\|T\| \leq \liminf_{n \rightarrow \infty} \|T_n\|$.

Proof. Fix $x \in X, T_n(x) \rightarrow T(x)$ so

$$\sup_n \|T_n(x)\| < +\infty$$

By Banach-Steinhaus theorem, $\sup_n \|T_n\| < +\infty$ so we have (1). Then note that

$$\|T(x)\| = \|\lim_{n \rightarrow \infty} T_n(x)\| = \lim_{n \rightarrow \infty} \|T_n(x)\| \leq \sup_n \|T_n\| \|x\|$$

$$\implies \|T\| \leq \sup_n \|T_n\|$$

Informally, $\{x_n\}$ is a sequence then $\liminf$ is the smallest possible limit of subsequence. If x_n is bounded from below then $\liminf x_n > -\infty$. Then

$$\begin{aligned} \lim_{n \rightarrow \infty} \|T_n(x)\| &= \liminf_{n \rightarrow \infty} \|T_n(x)\| \leq \liminf_{n \rightarrow \infty} \|T_n\| \|x\| \\ \implies \|T\| &\leq \liminf_{n \rightarrow \infty} \|T_n\| \end{aligned}$$

□

Corollary 6.5. The followings are equivalent (X -Banach, $A \subset X$):

- (1) A is bounded.
- (2) $\forall \varphi \in X^*, \sup_{a \in A} |\varphi(a)| < +\infty$.
- (3) $\forall \varphi \in X^*$, the set

$$\bigcup_{a \in A} \{\varphi(a)\}$$

is bounded.

Proof. ((1) $\implies$ (2)) If A is bounded then $\|a\| \leq C, \forall a \in A$.

$$\|\varphi(a)\| \leq \|\varphi\| \|a\| \leq C \|\varphi\| < \infty$$

((2) $\implies$ (3)) Obvious.

((2) $\implies$ (1)) $\forall x \in X$ we need to associated on X^* i.e.

$$\mathcal{J} : X \rightarrow (X^*)^*$$

$$(\mathcal{J}x)(\varphi) = \varphi(x) \in \mathbb{F}$$

If $\forall \varphi \in X^*, \sup_{a \in A} |\varphi(a)| < +\infty \implies \sup_{a \in A} \|a\| < \infty$. Then since $\varphi(a) = |(\mathcal{J}(a))(\varphi)|$ then by previous theorem we have the result required. □

Lemma 6.6. $\mathcal{J}$ is norm preserving but not always surjective.

Proof. By definition,

$$\begin{aligned} |(\mathcal{J}x)(\varphi)| &= |\varphi(x)| \leq \|\varphi\| \|x\|_X \\ \implies \|\mathcal{J}x\|_{X^{**}} &\leq \|x\|_X \end{aligned}$$

Recall lemma after Hahn-Banach theorem,

$$\|x\| = \max_{\varphi \in X^*, \|\varphi\|=1} |\varphi(x)|$$

Take φ such that $\|\varphi\| = 1$ and

$$|\varphi(x)| = \|x\|$$

then

$$\begin{aligned} |(\mathcal{J}x)(\varphi)| &= |\varphi(x)| = \|x\| \|\varphi\| \\ \implies \|\mathcal{J}x\| &\geq \|x\| \end{aligned}$$

□

Corollary 6.7. X -Banach, $A \subset X^*$ then followings are equivalent:

- (1) A is bounded,
- (2) $\forall x \in X, \sup_{\varphi \in A} |\varphi(x)| < +\infty$.
- (3) $\forall x \in X$, the set

$$\bigcup_{\varphi \in A} \{\varphi(x)\}$$

is bounded.

Definition 6.8. X, Y -metric space $T : X \rightarrow Y$ is open if $\forall U \subset X$ then $f(U)$ is open.

Example. Here are some examples of open map:

- (1) $f : X \rightarrow Y$ homeomorphism
- (2) $f : U \rightarrow U$ non-constant holomorphic.

Theorem 6.9. (Open mapping theorem) X, Y -Banach. $A \in \mathcal{L}(X, Y)$, if A is surjective then A is an open map.

Proof. (STEP 1) Sufficient to show that $\exists \varepsilon > 0$ such that $A(B_X(0, 1)) \supset B_Y(0, \varepsilon)$ then A open means that for any $U \subset X$ open, $\forall y \in A(U)$ one has $\exists \delta > 0, B(y, \delta) \subset A(U)$. First observe that U is open, so $\exists x \in U$ such that $y = Ax, B(x, \varepsilon') \subset U$. In particular, $A(U) \supset A(x) + A(B_X(0, \varepsilon'))$ by the given condition, $A(B_X(0, \varepsilon')) \supset B_Y(0, \varepsilon\varepsilon')$ so

$$A(U) \supset A(x) + B_Y(0, \varepsilon\varepsilon') = B_Y(y, \varepsilon\varepsilon')$$

(STEP 2) $\exists \varepsilon > 0$ such that $B_Y(0, \varepsilon) \subset \overline{A(B_X(0, 2))}$. We shall use BCT and the fact that Y is Banach. Since A is surjective, so

$$Y = \bigcup_n A(B_X(0, n))$$

$\exists n$ such that $\text{int} \overline{A(B_X(0, n))}$ is non empty. Therefore,

$$\text{int} \overline{A(B_X(0, 1))} \neq \emptyset \implies$$

$\exists y_0$ and $\varepsilon > 0$ such that

$$B_Y(y_0, \varepsilon) \subset \overline{A(B_X(0, 1))}$$

So we claim that

$$B_Y(0, \varepsilon) \subset \overline{A(B_X(0, 1))} - y_0 \subset \overline{A(B_X(0, 1)) - A(B_X(0, 1))} \subset \overline{A(B_X(0, 2))}$$

Indeed, take $y \in \overline{A(B_X(0, 1))} \implies \exists x_n \in B_X(0, 1)$ such that $Ax_n \rightarrow y$. $\exists z_n \in B_X(0, 1)$ such that $Az_n \rightarrow y_0$ then

$$A(x_n - z_n) \rightarrow y - y_0$$

$$\|x_n - z_n\| \leq \|x_n\| + \|z_n\| < 2$$

$$\implies y - y_0 \in \overline{A(B_X(0, 2))}$$

(STEP 3) $\exists \varepsilon > 0$ such that $B_X(0, \varepsilon) \subset A(B_X(0, 1))$. Since $B_Y(0, \varepsilon) \subset \overline{A(B_X(0, 2))}$ then

$$B_Y(0, 2^{-n}\varepsilon) \subset \overline{A(B_X(0, 2^{-n+1}))}$$

Take $y \in B_Y(0, \frac{\varepsilon}{2})$. We want to show $\exists x \in B_X(0, 1)$ such that $Ax = y$,

$$B_Y(0, \frac{\varepsilon}{2}) \subset \overline{A(B_X(0, \frac{1}{4}))}$$

$\exists x_1$ such that $\|x_1\| < \frac{1}{4}$ and

$$\|y - Ax_1\| < \frac{\varepsilon}{16}$$

then

$$B_Y(0, \frac{\varepsilon}{16}) \subset \overline{A(B_X(0, \frac{1}{8}))}$$

similarly $\exists x_2$ such that $\|x_2\| < \frac{1}{8}$ then

$$\|y - Ax_1 - Ax_2\| < \frac{\varepsilon}{32}$$

repeat this we have

$$\|y - (Ax_1 + \dots + Ax_n)\| < \frac{\varepsilon}{2^{n+4}}$$

Define $x = \sum_{n=1}^{\infty} x_n$, note that

$$\|x\| = \sum_{n=1}^{\infty} \frac{1}{2^{1+n}} = \frac{1}{2} < 1 \implies \|x\| < 1$$

and then $Ax = y$. □

Corollary 6.10. If $A \in \mathcal{L}(X, Y)$ is bijective then A^{-1} is bounded.

Corollary 6.11. If X is equipped with two norms $\|\cdot\|_1, \|\cdot\|_2$. Assume that X is complete with respect to $\|\cdot\|_1, \|\cdot\|_2$ and

$$\|x\|_2 \leq C\|x\|_1 \quad \forall x \in X$$

then $\|\cdot\|_1$ and $\|\cdot\|_2$ are equivalent.

Proof. Define $X_1 = (X, \|\cdot\|_1)$, $X_2 = (X, \|\cdot\|_2)$. Then there exists two maps $id_{x_1, x_2} : x_1 \rightarrow x_2$, $id_{x_2, x_1} : x_2 \rightarrow x_1$. Thus, since $\|x\|_2 \leq C\|x\|_1$,

$$\begin{aligned} \|id_{x_1, x_2}(x)\|_2 &\leq C\|x\|_1 \\ \implies id_{x_1, x_2} &\in \mathcal{L}(X_1, X_2) \end{aligned}$$

By open mapping theorem,

$$\begin{aligned} (id_{x_1, x_2})^{-1} &\in \mathcal{L}(X_1, X_2) \\ \implies \|x\|_1 &\leq C'\|x\|_2 \end{aligned}$$
□

Definition 6.12. $A : X \rightarrow Y$ be a map, graph of A is

$$\Gamma(A) = \{(x, Ax) | x \in X\}$$

Theorem 6.13. (Closed graph theorem) X, Y —two Banach space then

$$X \oplus Y = X \times Y$$

$$\|(x, y)\|_{X \times Y} = \max(\|x\|, \|y\|)$$

if A is a linear operator, $\Gamma(A)$ is closed then $A \in \mathcal{L}(X, Y)$.

Proof. A -linear implies $\Gamma(A)$ is a subspace in $X \oplus Y$ which implies $\Gamma(A)$ is Banach. Let $P : \Gamma(A) \rightarrow x$, $P(x, Ax) = x$ which is bounded. In addition, P is bijective so $P^{-1} : X \rightarrow \Gamma(A)$ is bounded. $\|Ax\| \leq \max(\|x\|, \|Ax\|) = \|(x, Ax)\|_{P^{-1}(x)}$ which is bounded. $\square$

7 Weak convergence

Definition 7.1. X -normed space. We say that $\{x_n\}$ converges to x weakly if $\forall \varphi \in X^*$ one has $\varphi(x_n) \rightarrow \varphi(x)$ then $x_n \rightharpoonup x$.

Theorem 7.2. $\{x_n\} \subset X, x \in X$ then

- (1) If $x_n \rightarrow x$, then $x_n \rightharpoonup x$.
- (2) If $x_n \rightharpoonup x$, then $\{x_n\}$ is bounded and $\|x\| \leq \liminf_{n \rightarrow \infty} \|x_n\|$.
- (3) $x_n \rightharpoonup x$ and $\varphi_n \rightarrow \varphi$ in X^* then $\varphi_n(x_n) \rightarrow \varphi(x)$.
- (4) Weak limits are unique. i.e. if $x_n \rightharpoonup x$ and $x_n \rightharpoonup y$ then $x = y$.

Proof.

- (1) We know $\|x_n - x\| \rightarrow 0$ we want to show $\varphi \in X^* \implies \varphi(x_n) \rightarrow \varphi(x)$. Note that

$$|\varphi(x_n) - \varphi(x)| = |\varphi(x - x_n)| \leq \|\varphi\| \|x_n - x\| \rightarrow 0$$

- (2) $x_n \rightharpoonup x \implies \{x_n\}$ is weakly bounded. i.e. $\varphi(x_n)$ is converging sequence so

$$\bigcup_n \{\varphi(x_n)\}$$

is a bounded set. Recall from corollary from Hahn-Banach theorem, $\{x_n\}$ is bounded. Define $J : X \rightarrow X^{**}$ where $(J(x))(\varphi) = \varphi(x)$ and $\|J(x)\| = \|x\|$. Look $J(x_n)$ note that $x_n \rightharpoonup x \Leftrightarrow \forall \varphi \in X^*, \varphi(x_n) \rightarrow \varphi(x) \Leftrightarrow [J(x_n)](\varphi) \rightarrow [J(x)](\varphi) \Leftrightarrow J(x_n) \rightarrow J(x)$ pointwise as elements of $\mathcal{L}(X^*, \mathbb{F})$ so

$$\begin{aligned} \|J(x)\| &\leq \liminf \|J(x_n)\| \\ &\implies \|x\| \leq \liminf \|x_n\| \end{aligned}$$

- (3) $|\varphi(x) - \varphi_n(x_n)| \leq |\varphi(x) - \varphi(x_n)| + |\varphi(x_n) - \varphi_n(x_n)| \leq |\varphi(x) - \varphi(x_n)| + \|\varphi - \varphi_n\| \|x_n\| \rightarrow 0$ for large enough n .
- (4) Suppose $x_n \rightharpoonup y$ and $x_n \rightharpoonup x$, suppose $x - y \neq 0$ then $\exists \varphi \in X^*$ such that $\varphi(x - y) \neq 0$ by Hahn-Banach theorem. Then

$$\varphi(x_n) \rightarrow \varphi(y), \quad \varphi(x_n) \rightarrow \varphi(x)$$

but $\varphi(x - y) \neq 0$.

□

Examples.

- (1) $X = \ell^p, 1 < p < \infty$ and $e^{(n)} = (0, \dots, 1, 0, \dots)$. Then $e^{(n)} \rightharpoonup 0$
- (2) $X = \ell^1$ then $e^{(n)}$ does not converge weakly.

Proof.

- (1) Take $\varphi \in (\ell^p)^*$ then $\exists y \in \ell^q$ such that

$$\begin{aligned} \varphi(x) &= \sum x_i y_i \\ \varphi(e^{(n)}) &= y_n \rightarrow 0 \end{aligned}$$

- (2) Take φ such that

$$\varphi(e^{(n)}) = \begin{cases} 0 & n \text{ is odd} \\ 1 & n \text{ is even} \end{cases}$$

Indeed,

$$\varphi(x) = \sum_{i=0}^{\infty} x_{2i+1}$$

then $\varphi(e^{(n)})$ does not converge.

□

Proposition 7.3. If $\dim(X) < +\infty$ then $x_n \rightharpoonup x$ implies $x_n \rightarrow x$.

Proof. Choosing $e_1, \dots, e_d$ be basis in X where

$$x_n = \sum a_n^i e_i, \quad x = \sum a^i e_i$$

Define $\varphi_i \in X^*$ to be such that

$$\begin{aligned} \varphi_i(\sum a^i e_i) &= a_i \\ x_n \rightharpoonup x &\implies \varphi_i(x_n) \rightarrow \varphi_i(x) \implies a_n^i \rightarrow a^i \\ &\implies \|x_n - x\|_{\infty} \rightarrow 0 \implies \|x_n - x\| \rightarrow 0 \end{aligned}$$

since all norms on X are equivalent.

□

Definition 7.4. (Weak-* convergence) $\{\varphi_n\} \subset X^*$, $\varphi \in X^*$. We say φ_n converges weakly-* to φ if $\forall x \in X$,

$$\varphi_n(x) \rightarrow \varphi(x), \quad \varphi_n \rightharpoonup^* \varphi$$

let $J : X \rightarrow X^{**}$ be norm-preserving then this is equivalent to

$$[J(x)](\varphi_n) \rightarrow [J(x)](\varphi)$$

Proposition 7.5. $\{\varphi_n\} \subset X^*$, $\varphi \in X^*$ then

- (1) $\varphi_n \rightharpoonup \varphi \implies \varphi_n \rightharpoonup^* \varphi$.
- (2) $\varphi_n \rightharpoonup^* \varphi \implies \{\varphi_n\}$ is bounded and $\|\varphi\| \leq \liminf \|\varphi_n\|$.
- (3) $\varphi_n \rightharpoonup^* \varphi$, $x_n \rightarrow x \implies \varphi_n(x_n) \rightarrow \varphi(x)$.
- (4) If $\varphi_n \rightharpoonup^* \varphi$ and $\varphi_n \rightharpoonup^* \psi$ then $\varphi = \psi$.

Example. (1) $1 \leq p < +\infty$, $\varphi_n \in (\ell_p)^*$ where

$$\varphi_n(x) = x_n$$

then $\varphi_n \rightharpoonup^* 0$

Proof. Want to show $\varphi_n(x) \rightarrow 0$, $\forall x \in \ell^p$. This is true since $x_n \rightarrow 0$ for $p < +\infty$. □

(2) $\varphi_n(x) = x_n$, $\varphi_n \in (c_0)^*$ then $\varphi \rightharpoonup^* 0$ but φ_n do not converge weakly.

Proof. $\varphi_n \rightharpoonup^* 0$ for the same reason previously. Since $(c_0)^* \cong \ell^1$ so the weak convergence of φ_n is equivalent to the weak convergence

$$\varphi_n \rightharpoonup e^{(n)}$$

in ℓ^1 which does not converge weakly. □

Theorem 7.6. X -separable and Banach space. Then $\exists$ a metric ρ on B_{X^*} such that $\rho(\varphi_n, \varphi) \rightarrow 0$ if and only if $\varphi_n \rightharpoonup^* \varphi$.

Proof. Since X is separable, let $\{x_n\}$ is a countable dense set in the unit ball B_X . The idea is $\forall x \in X$, $\varphi_n(x) \rightarrow \varphi(x) \iff \forall k, \varphi_n(x_k) \rightarrow \varphi(x_k)$. Define

$$\rho(\varphi, \psi) = \sum_{k=1}^{\infty} \frac{|\varphi(x_k) - \psi(x_k)|}{2^k}$$

Suppose that $\varphi_n \rightharpoonup^* \varphi$ so we want to show $\rho(\varphi_n, \varphi) \rightarrow 0$.

$$|\varphi_n(x_k) - \varphi(x_k)| \leq \|\varphi_n\| \|x_k\| + \|\varphi\| \|x_k\| \leq 2$$

$$\varphi_n(x_k) \rightarrow \varphi(x_k)$$

then

$$\rho(\varphi_n, \varphi) = \sum_{k=1}^K \frac{|\varphi_n(x_k) - \varphi(x_k)|}{2^k} + \sum_{k=K+1}^{\infty} \frac{|\varphi_n(x_k) - \varphi(x_k)|}{2^k}$$

Choosing K so that $2^{-K+1} < \frac{\varepsilon}{2}$ and choose N such that $\forall n > N$,

$$|\varphi_n(x_k) - \varphi(x_k)| < \varepsilon \quad \forall k = 1, \dots, K$$

$$\implies \rho(\varphi_n, \varphi) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Now we show the opposite. Suppose $\rho(\varphi_n, \varphi) \rightarrow 0$ and we want to show $\forall y \in X$, $\varphi_n(y) \rightarrow \varphi(y)$. $\exists x_{k_j}$ such that $x_{k_j} \rightarrow \frac{y}{\|y\|} \in B_X$.

$$\rho(\varphi_n, \varphi) = \sum_{k=1}^{\infty} \frac{|\varphi_n(x_k) - \varphi(x_k)|}{2^k}$$

In particular,

$$|\varphi_n(x_{k_j}) - \varphi(x_{k_j})| \leq 2^{k_j} \rho(\varphi_n, \varphi)$$

Choose j such that

$$\|x_{k_j} - \frac{y}{\|y\|}\| < \delta$$

$$|\varphi_n(y) - \varphi(y)| = \|\varphi_n(\frac{y}{\|y\|}) - \varphi(\frac{y}{\|y\|})\|$$

$$|\varphi_n(\frac{y}{\|y\|}) - \varphi(\frac{y}{\|y\|})| \leq |\varphi_n(x_{k_j}) - \varphi(x_{k_j})| + |\varphi_n(x_{k_j}) - \varphi_n(\frac{y}{\|y\|})| + |\varphi(x_{k_j}) - \varphi(\frac{y}{\|y\|})|$$

$$\leq 2^{k_j} \rho(\varphi_n, \varphi) + (\|x_{k_j} - \frac{y}{\|y\|}\|)(\|\varphi\| + \|\varphi_n\|)$$

we can choose j to make it small. □

Theorem 7.7. X -separable and Banach. Then $\forall \{\varphi_n\} \subset B_{X^*}$, $\exists$ convergent subsequence.

Proof. Given x_1 then $\{\varphi_n(x_1)\}_n$ be bounded set

$$|\varphi_n(x_1)| \leq \|\varphi_n\| \|x_1\| \leq \|x_1\|$$

then there exists convergent subsequence

$$\varphi_{n_1}^{(1)}(x_1), \varphi_{n_2}^{(1)}(x_1), \dots, \{n_j\} \subset \Lambda_1 \subset \mathbb{N}$$

$$\varphi_{n_1}^{(2)}(x_1), \varphi_{n_2}^{(2)}(x_1), \dots, \{n_j\} \subset \Lambda_2 \subset \mathbb{N}$$

Instead, the k th element of Λ is the k th element of $\Lambda_k \implies$ all elements in Λ are k th element in Λ_k . Then

$$\exists \lim_{n \in \Lambda} \varphi_n(x_j), \quad \forall j = 1, \dots, k$$

We define

$$\varphi(x_k) = \lim_{n \rightarrow \infty} \varphi_n(x_k) \quad \text{on } \text{span}\{x_k\}$$

By continuity, we extend φ to a functional on $\overline{\text{span}\{x_k\}} = X$ if $y \in X$, $\exists x_k \rightarrow y$ such that

$$\begin{aligned} |\varphi_n(y) - \varphi(y)| &< \varepsilon \\ &\leq |\varphi_n(y - x_k)| + |\varphi(y - x_k)| + |\varphi_n(x_k) - \varphi(x_k)| \\ &\leq (\|\varphi_n\| + \|\varphi\|)(\|y - x_k\|) + |\varphi_n(x_k) - \varphi(x_k)| \end{aligned}$$

□

Theorem 7.8. X -separable and Banach. Then the unit ball B_{X^*} in X^* is sequentially compact.

Definition 7.9. X is called reflexive if $J : X \rightarrow X^{**}$ is surjective.

Examples. (1) Any finite dimensional spaces are reflexive, i.e. $\dim X = \dim X^* = \dim X^{**}$.

(2) ℓ^p , $1 < p < +\infty$ then if $\frac{1}{p} + \frac{1}{q} = 1$ then

$$(\ell^p)^{**} \cong (\ell^q)^* \cong \ell^p$$

we shall prove ℓ^p is reflexive.

Proof. Define $J_p : \ell^p \rightarrow (\ell^p)^{**}$, $\Psi_q : \ell_q \rightarrow (\ell_p)^*$, $\Psi'_q : (\ell^p)^{**} \rightarrow (\ell^q)^*$, $\Psi_p : \ell^p \rightarrow (\ell^q)^*$. We claim that $\Psi'_q J_p = \Psi_p$ i.e. J_p is an isomorphism.

$$[\Psi'_q J_p(x)](y) = [J_p](\Psi_q y) = (\Psi_q y)(x) = \sum_{i=1}^{\infty} x_i y_i = (\Psi_p(x))(y)$$

□

(3) ℓ^1 and ℓ^∞ are not reflexive.

(4) If H is Hilbert space then H is reflexive.

Proof. Define $I_H : H \rightarrow H^*$ which is conjugate linear then

$$[I_H(x)](y) := (y, x)$$

define $I'_H : H^{**} \rightarrow H^*$ and $J_H : H \rightarrow H^{**}$ and we claim that $I'_H J_H = I_H$.

□

Definition 7.10. $A : X \rightarrow Y$, $A \in \mathcal{L}(X, Y)$ define $A' : Y^* \rightarrow X^*$ by

$$\mathbb{F} \ni (A'\varphi)(x) := \varphi(Ax)$$

which is equivalent to $\langle A'\varphi, x \rangle = \langle \varphi, Ax \rangle$.

Lemma 7.11.

- (1) A' is unique, well-defined and $id'_X = id_{X^*}$.
- (2) $A : X \rightarrow Y$, $B : Y \rightarrow X$ then $(BA)' = A'B'$.
- (3) If A is an isomorphism then A' is an isomorphism.
- (4) $\|A'\| \leq \|A\|$

Proof.

(1),(4) $|(A'\varphi)(x)| = |\varphi(Ax)| \leq \|\varphi\| \cdot \|Ax\| \leq \|\varphi\| \|A\| \|x\|$. Therefore, $\|A'\varphi\| \leq \|\varphi\| \|A\| \implies A'\varphi \in X^* \implies \|A'\| \leq \|A\|$. By definition, $id_X : X \rightarrow X$, $id : X^* \rightarrow X^*$ and

$$\begin{aligned} [id'_X(\varphi)](x) &= \varphi(id_X(x)) = \varphi(x) \\ \implies id'_X(\varphi) &= \varphi \implies id'_X = id_{X^*} \end{aligned}$$

(2) Define $A : X \rightarrow Y$, $B : Y \rightarrow Z$ and $A' : Y^* \rightarrow X^*$, $B' : Z^* \rightarrow Y^*$. Then

$$\begin{aligned} [(BA)'(\varphi)](x) &:= \varphi(BAx) \\ [A'B'(\varphi)](x) &= [B'\varphi](Ax) = \varphi(BAx) \end{aligned}$$

(3) If A is an isomorphism then $\exists A^{-1} : Y \rightarrow X$ such that $AA^{-1} = id_Y$ and $A^{-1}A = id_X$. Take the dual we have

$$(AA^{-1})' = id_{Y^*}, \quad (A^{-1}A)' = id_{X^*}$$

so A' is an isomorphism. □

Proposition 7.12. X is Banach then X is reflexive if and only if X^* is reflexive.

Proof. Define $J_X : X \rightarrow X^{**}$, $J_{X^*} : X^* \rightarrow X^{***}$, $J'_X : X^{***} \rightarrow X^*$. Then we claim that $J'_X \circ J_{X^*} = id_{X^*}$ then

$$[(J'_X \circ J_{X^*})(\varphi)](x) = [J_{X^*}(\varphi)](J_X(x)) = [J_X(x)](\varphi) = \varphi(x)$$
□

Proposition 7.13. Suppose X is Banach, if X^* is separable then X is separable.

Proof. Suppose X^* is separable then $V = \{\varphi \in X^*, \|\varphi\| = 1\}$ is also separable then $\{\varphi_n\}_{n=1}^\infty$ is dense in V with $\|\varphi_n\| = 1$. Then $\exists x_n \in X$, such that $\|x_n\| = 1$ and $\varphi_n(x_n) > \frac{1}{2}$. We claim that

$$X = \overline{\text{Span}(\{x_n\})}$$

Suppose $L = \overline{\text{Span}(\{x_n\})} \neq X$ then $\exists \varphi \in X^*$ such that $\|\varphi\| = 1$ and $\varphi(L) = 0$ i.e. $L \subset \ker \varphi$ so $\varphi(x_n) = 0, \forall n$. But since φ_n is dense so $\exists n$ such that $\|\varphi - \varphi_n\| < \frac{1}{2}$ then

$$0 = \varphi(x_n) = \varphi_n(x_n) - (\varphi_n(x_n) - \varphi(x_n))$$

Note that $|(\varphi - \varphi_n)(x_n)| \leq \|\varphi - \varphi_n\| \|x_n\| \leq \frac{1}{2}$ then

$$0 = |\varphi(x_n)| \geq |\varphi_n(x_n)| - |\varphi_n(x_n) - \varphi(x_n)| \geq \frac{1}{2} - \frac{1}{2} = 0$$

so we reach a contradiction. □

Proposition 7.14. X is separable and reflexive if and only if X^* is separable and reflexive.

Proof. ($\Leftarrow$) X^* is separable implies X is separable.

($\Rightarrow$) $J_X : X \rightarrow X^{**}$ is isomorphism then X^{**} is separable so X^* is separable. □

Corollary 7.15. ℓ^1 and ℓ^∞ are not reflexive.

Proof. Since $(\ell^1)^* \cong \ell^\infty$ then it suffices to show that ℓ^1 is not reflexive. If ℓ^1 is reflexive and since ℓ^1 is separable so $(\ell^1)^*$ is separable and reflexive so ℓ^∞ is separable which reach a contradiction. □

Theorem 7.16. If X is Banach and separable then any sequence $\{\varphi_n\} \subset B_{X^*}$ has weak* converging subsequence.

Corollary 7.17. If X is Banach and separable and reflexive then any sequence $\{\varphi_n\} \subset B_{X^*}$ has weakly converging subsequence.

Proof. Let $J : X \rightarrow X^{**}$ be an isomorphism and $\{J(x_n)\} \subset X^{**}$, suppose $J(x_n)$ converges weakly* to Ψ then

$$[J(x_n)](\varphi) \rightarrow \Psi(\varphi)$$

so $\exists x$ such that $\Psi = J(x) \implies x_n \rightharpoonup x$. $\|J(x_n)\| \leq \|x_n\| \leq 1$ and $\{J(x_n)\} \subset B_{X^{**}}$ then by the theorem above we know there exists weak* converging subsequence so there exists weakly converging subsequence. □

Proposition 7.18. The polynomial

$$P(x) = x^{2k+1} + a_1 x^{2k} + \cdots + a_{2k+1}$$

has a real root.

Proof. Let $P(x) = Q'(x)$ then $Q(x) = \frac{x^{2k+2}}{2k+2} + \cdots$ then $P(x_0) = 0$ if

$$Q(x) = \inf_{x \in \mathbb{R}} Q(x) = m$$

Note that $\exists C, M$

$$Q(x) \geq C|x|^{2k+2} - M$$

so $m > -\infty$ since $Q(x) \geq -M$. Then we also have

$$\{x | Q(x) \leq m + 1\} \neq \emptyset$$

so

$$\begin{aligned} C|x|^{2k+2} - M \leq Q(x) \leq m + 1 &\implies |x|^{2k+2} \leq \frac{m + 1 + M}{C} \\ &\implies \{x | Q(x) \leq m + 1\} \subset \text{some ball} \end{aligned}$$

In addition, $\exists x_n$ such that $Q(x_n) \rightarrow m$ then

$$\{x_n\} \subset \{x | Q(x) \leq m+1\} \subset \text{ball}$$

since the ball is compact so $\exists$ some subsequence $x_{n_k} \rightarrow x_0$ i.e. $Q(x_0) = m$. □

Example. Consider

$$\begin{aligned} -\nabla u &= f \text{ in } \Omega \\ u &= 0 \text{ on } \partial\Omega \end{aligned}$$

then $\forall v \in C_0^\infty(\Omega)$

$$\int_{\Omega} -\nabla \cdot v \, dx = \int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx$$

if u is the weak solution then u solves the solution. We claim that suppose u_0 is such that

$$\begin{aligned} F[u] &= \frac{1}{2} \int_{\Omega} |\nabla u|^2 \, dx - \int_{\Omega} f u \, dx \\ \inf_{u|_{\partial\Omega}=0} F(u) &= F[u_0] \end{aligned}$$

then it is a weak solution.

Proof. Note that

$$F[u_0 + tv] = \frac{1}{2} \int_{\Omega} |\nabla(u_0 + tv)|^2 \, dx - \int_{\Omega} f(u_0 + tv) \, dx$$

then

$$\frac{d}{dt} \Big|_{t=0} F[u_0 + tv] = \int_{\Omega} \langle \nabla u_0, \nabla v \rangle \, dx - \int_{\Omega} f v \, dx = 0$$

□

We only need to find some m such that

$$\inf_{u|_{\partial\Omega}=0} F(u) = m$$

Define the norm on $H_0^1(\Omega)$ by

$$\|u\|_{H_0^1}^2 = \int_{\Omega} |\nabla u|^2 \, dx + \int_{\Omega} |u|^2 \, dx$$

H_0^1 is the completion of C_0^∞ with respect to $\|\cdot\|_{H^1}$. The fact is that

$$\|\cdot\|_{H_0^1}^2 = \int_{\Omega} |\nabla u|^2 \, dx$$

then

$$F[u] = \frac{1}{2} \int_{\Omega} |\nabla u|^2 \, dx - \int_{\Omega} f u \, dx$$

then

$$\begin{aligned} \frac{1}{2} \int_{\Omega} |\nabla u|^2 \, dx &\geq C \|u\|_{H^1}^2, \quad \int_{\Omega} f u \, dx \leq \|f\|_{L^2} \|u\|_{H_0^1} \leq \frac{C}{2} \|u\|_{H_0^1}^2 + \frac{1}{2C} \|f\|_{L^2}^2 \\ \implies F[u] &\geq \frac{C}{2} \|u\|_{H_0^1}^2 - \frac{2}{2C} \|f\|_{L^2}^2 \end{aligned}$$

then

$$m = \inf_{u \in H_0^1} F(u) > -\infty$$

and when we look at $\{u | F(u) \leq m+1\}$ then

$$\|u\|_{H_0^1} \leq B$$

in addition, $\exists u_n$ such that $F(u_n) \rightarrow m$. But we don't know $u_n \rightarrow u$ since the balls in H_0^1 is not compact. But there exists $u_{n_k} \rightarrow u_0$ since H_0^1 is Hilbert space so reflexive. Then

$$F(u_{n_k}) = \frac{1}{2} \|u_{n_k}\|_{H_0^1}^2 - \int_{\Omega} f u_{n_k} \, dx$$

Since $\lim_{k \rightarrow \infty} \|u_{n_k}\| \leq \|u_0\|$

$$\begin{aligned} F(u_0) &\leq \lim_{k \rightarrow \infty} F(u_{n_k}) = m = \inf_u F(u) \\ \implies F(u_0) &= m \end{aligned}$$

8 Compact operators and the Fredholm alternative

Theorem 8.1. Denote $R(L) = \text{im } L$, let H be a Hilbert space with $\dim H < +\infty$ and $L \in \mathcal{L}(H)$ consider

$$Lx = y$$

then we have

- (1) $Lx = y$ has a unique solution $\forall y \in H$.
- (2) $Lx = y$ has a solution only if

$$y = R(L) = \ker(L^*)^\perp$$

Proposition 8.2. H is a Hilbert space, then

$$\overline{R(L)} = \ker(L^*)^\perp$$

Proof. Note that $R(L) \subset \ker(L^*)^\perp$, since $y = Lx \in R(L)$, pick $z \in \ker(L^*)^\perp$ then

$$(y, z) = (Lx, z) = (x, L^*z) = 0$$

then we would like to show that $R(L)^\perp \subset \ker(L^*)$, $z \in R(L)^\perp$ so

$$(Lx, z) = 0 \quad \forall x$$

$$\implies 0 = (Lx, z) = (x, L^*z)$$

if we take $x = L^*z$ then

$$(L^*z, L^*z) = 0 \implies L^*z = 0 \implies z \in \ker(L^*)$$

□

Remark. For general $L \in \mathcal{L}(H)$ if $\dim H = +\infty$ then $R(L)$ is not closed. The operators for which this then works as stated are called "Fredholm".

Definition 8.3. Let $A : H \rightarrow H$, $A \in \mathcal{L}(H)$. A is compact if $\overline{A(B(0, 1))}$ is compact.

Proposition 8.4. Properties of compact operators: $A, C \in \mathcal{L}(H)$,

- (1) if at least one of them compact then CA is compact.
- (2) if both are compact then $A + C$ is compact.

Proof.

- (1) WLOG say A is compact then $\overline{A(B(0, 1))}$ is compact so $C(\overline{A(B(0, 1))})$ is compact. Note that

$$\overline{CA(B(0, 1))} \subset C(\overline{A(B(0, 1))})$$

so $\overline{CA(B(0, 1))}$ is compact since it's closed subset of compact set. If C is compact note that

$$A(B(0, 1)) \subset B(0, R) \implies C(A(B(0, 1))) \subset C(B(0, R)) = RC(B(0, 1))$$

$$\implies \overline{C(A(B(0, 1)))} \subset \overline{RC(B(0, 1))}$$

- (2) If A, C are both compact then

$$\overline{(A + C)(B(0, 1))} \subset \overline{A(B(0, 1))} + \overline{C(B(0, 1))}$$

which is compact.

□

Theorem 8.5. (Fredholm alternative) H -Hilbert space, $A \in \mathcal{L}(H)$, A is compact and $\lambda \in \mathbb{C} \setminus \{0\}$ then $R(A - \lambda Id)$ is closed if one of the following holds

- (1) $\exists$ solution to $Ax = \lambda x$ and the dimension of solution space is finite.
- (2) $A - \lambda Id$ is invertible.

Example. $H = L^2[0, 1]$ then $(Mf)(x) = xf(x)$ is not compact.

Suppose the inverse, then either

- (1) $M - Id$ is invertible or
- (2) $\exists f \neq 0$ such that $(M - Id)f = 0$.

On one hand suppose $f \neq 0$ such that $(M - Id)f = 0$ then $(x - 1)f = 0 \implies f = 0$. On the other hand, $(x - 1)f = 1 \in L^2[0, 1]$ but $f = \frac{1}{x-1}$ is not in $L^2[0, 1]$.

Lemma 8.6. Let $\Phi = A - Id$ then $\dim \ker(\Phi) < +\infty$.

Proof. $\ker(\Phi)$ is closed subspace in H denote the subspace by L and let $A = Id_L$ then

$$\overline{A(B_L(0,1))} \subset \overline{B_L(0,1)} = B_L[0,1]$$

If $\dim L = +\infty$ then $B_L[0,1]$ is not compact but

$$B_L[0,1] \subset \overline{A(B_H(0,1))}$$

implies $B_L[0,1]$ is compact. □

Proposition 8.7. $R(\Phi)$ is closed.

Proof. Let $\{y_n\} = \{\Phi x_n\} \subset R(\Phi)$ such that $y_n \rightarrow y$ then we want to show $\exists x$ such that $y = \Phi(x)$. Let $x_n = s_n + z_n$ where $s_n \in \ker(\Phi)$ and $z_n \in \ker(\Phi)^\perp$ so

$$y_n = \Phi(x_n) = \Phi(z_n)$$

We hope $\{z_n\}$ converge. We first show $\{z_n\}$ is bounded. Suppose not, then up to a subsequence

$$\|z_n\| \rightarrow +\infty$$

define $\tilde{z}_n = \frac{z_n}{\|z_n\|}$ then $\|\tilde{z}_n\| = 1$. If A is compact up to a subsequence then $A\tilde{z}_n \rightarrow \tilde{w}$. Note that

$$\tilde{z} = -\Phi(\tilde{z}_n) + A(\tilde{z}_n) \rightarrow \tilde{w}$$

$$\Phi(\tilde{z}_n) = \frac{\Phi(z_n)}{\|z_n\|} \rightarrow 0$$

so $\Phi(\tilde{z}_n) \rightarrow 0 = \Phi(\tilde{w}) \in \ker(\Phi) \implies \tilde{w} \subseteq \ker(\Phi)$ so $\tilde{w} = 0$ contradicts with $\|\tilde{w}\| = 1$. Up to a subsequence, $Az_n \rightarrow w$ and

$$z_n = -\Phi(z_n) + A(z_n) \rightarrow w - y$$

$\Phi(z_n) \rightarrow y \implies y \in R(\Phi)$ □

Lemma 8.8. $\exists k \in \mathbb{N}$ such that

$$\begin{aligned} R(\Phi^j) &= R(\Phi^k) \quad \forall j \geq k \\ \ker(\Phi^j) &= \ker(\Phi^k) \quad \forall j \geq k \end{aligned}$$

Proof. Note

$$\Phi^k = (A - Id)^k = (\dots + (-1)^k Id)$$

so $R(\Phi^k)$ is closed and $\ker(\Phi^k)$ is finite dimensional. $\ker(\Phi) \subset \ker(\Phi^2) \subset \dots \subset \ker(\Phi^k) \subset \dots$. Assume it does not stabilise, i.e. $\exists n_j \rightarrow +\infty$ such that

$$\ker(\Phi^{n_j}) \subsetneq \ker(\Phi^{n_j+1})$$

Take $y_j \in \ker(\Phi^{n_j})$ and assume that $\|y_j\| = 1, \forall j \neq j$ then

$$\|Ay_k - Ay_j\| = \|(Id + \Phi)_{y_k} - (Id + \Phi)_{y_j}\|$$

$x \in \ker(\Phi^N) \implies \Phi_x \in \ker(\Phi^{N-1})$ so

$$\|(Id + \Phi)_{y_k} - (Id + \Phi)_{y_j}\| = \|-y_j + (\dots \in y_j^\perp)\| \geq 1$$

□

Lemma 8.9. Any element of H can be written as a sum

$$x = y + z$$

where $y \in R(\Phi^k)$ and $z \in \ker(\Phi^k)$ in a unique way.